Radial transfer estimates and polygon length laws for honeycomb walks

Abstract

A self-avoiding polygon on the honeycomb lattice is a closed loop that never visits a vertex twice. Give a polygon the critical weight $w(\gamma) = x_*^{|\gamma|}$, where $|\gamma|$ is its number of edges and $x_* = (2+\sqrt{2})^{-1/2}$ is the honeycomb connective constant raised to the power -1. We prove that the total critical weight of polygons of diameter at least r is $r^{-2+o(1)}$, that the same sum carrying one power of the length is $r^{-2/3+o(1)}$, and that the truncated second length moment is at most $r^{2/3+o(1)}$. As a consequence a polygon sampled at criticality and conditioned to have length at least n has diameter $n^{3/4+o(1)}$ with probability tending to one. The exponent $3/4$ is the long-predicted size exponent of planar self-avoiding walk; the proof here arrives at it through positive polygonal estimates and does not assume a scaling limit.

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1 Introduction: what is being counted, and what is attacked

1.1 The game

Place vertices at the points $P_{ij} = ih + jv$ and $R_{ij} = P_{ij} + (h + v)/3$ of the plane, where

$$h = (\sqrt{3}, 0), \qquad v = (\sqrt{3}/2, 3/2), \qquad i, j \in \mathbb{Z}.$$

Every P_{ij} is joined to the three R-vertices $R_{ij}, R_{i-1,j}, R_{i,j-1}$, and every edge has length one. This is the regular *honeycomb* (hexagonal) lattice; it is the lattice of a chicken-wire fence and of a single sheet of graphene.

A *self-avoiding polygon* γ is a finite cycle $p_1 p_2 \cdots p_L p_1$ of distinct lattice vertices, each consecutive pair joined by an edge. Equivalently it is a round walk that never crosses itself and never reuses a corner. We write $L = |\gamma|$ for its number of edges and $D = \operatorname{diam} \gamma$ for its Euclidean diameter, i.e. the largest distance between two of its points.

Polygons that differ only by a translate are the same object for us; we count each translation class once. This throws away the position and keeps the shape and size.

The critical weight of a class γ is

$$w(\gamma) = x_*^{|\gamma|}, \qquad x_* = (2 + \sqrt{2})^{-1/2}.$$

This particular number is not a coincidence: Duminil-Copin and Smirnov proved that on the honeycomb lattice the connective constant equals $\mu = \sqrt{2 + \sqrt{2}}$, i.e. the number of m-step self-avoiding walks grows like μ^m, so $x_* = 1/\mu$ is exactly the *critical* value at which the length series

$\sum_m (\#m\text{-step walks})\, x^m$ sits on the boundary between convergence and divergence. We work at exactly this value; it is the analog of the critical temperature of a magnet. Because x_* is the critical activity, the weights $w(\gamma)$ balance every possible length equally and the only thing left to measure is size.

1.2 The three observables

For an unrooted polygon (one considered up to translation) define, for $r > 0$ a length scale (thought of as large),

$$A(r) = \sum_{\operatorname{diam}\gamma \geq r} w(\gamma) \qquad \text{(mass of big-diameter polygons)}, \tag{1}$$

$$P(r) = \sum_{\operatorname{diam}\gamma \geq r} |\gamma|\, w(\gamma) \qquad \text{(the same with one length factor)}. \tag{2}$$

Also, summing over the complementary range (diameter below r) but carrying *two* length factors,

$$M_2(r) = \sum_{\operatorname{diam}\gamma \leq r} |\gamma|^2 w(\gamma) \qquad \text{(truncated second length moment)}. \tag{3}$$

All three are finite sums for each fixed r; we will prove they stay controlled as $r \to \infty$.

1.3 The main result

Write $f(r) = r^{a+o(1)}$ if for every $\varepsilon > 0$ one has $r^{a-\varepsilon} \leq f(r) \leq r^{a+\varepsilon}$ for all sufficiently large r. The one-sided notation $f(r) \leq r^{a+o(1)}$ is read the same way with two inequalities removed.

Theorem 1.1 (Diameter tails and second length moment)**.** *For the translation classes of honeycomb self-avoiding polygons with the critical weight $w(\gamma) = x_*^{|\gamma|}$,*

$$A(r) = r^{-2+o(1)}, \qquad P(r) = r^{-2/3+o(1)}, \qquad M_2(r) \leq r^{2/3+o(1)} \qquad (r \to \infty).$$

The first reading of the theorem: most of the critical weight sits on small polygons, and the mass above diameter r decays like r^{-2}. The second reading, dividing P by A, says that among big polygons the average length is $r^{4/3}$ (since $r^{-2/3}/r^{-2} = r^{4/3}$). Polygons conditioned on large diameter are long: the mean length is *larger* than the diameter, which is the geometric statement that in a good approximation the polygon looks like a fractal curve that wanders through its own bounding box repeatedly.

The second moment M_2 controls the opposite failure mode: it rules out the critical mass being concentrated on very long polygons that hug a tiny box. Together the three lines in Theorem 1.1 squeeze the size law from both sides.

1.4 The consequence for length cutoffs

Condition now on the length of the polygon instead of its diameter. The long-standing prediction of the polymer literature (Nienhuis; Lawler, Schramm and Werner) is that a critical polygon of length n has typical diameter of order $n^{3/4}$. The next corollary turns Theorem 1.1 into exactly that.

Corollary 1.2 (Length cutoffs, typical size 3/4)**.** *The length tails are*

$$\sum_{L(\gamma)\geq n} w(\gamma) = n^{-3/2+o(1)}, \qquad \sum_{L(\gamma)\geq n} L(\gamma)\, w(\gamma) = n^{-1/2+o(1)}.$$

Under either normalized weight $w(\gamma)$ or $L(\gamma)w(\gamma)$, conditioned on $L(\gamma) \geq n$, one has $\operatorname{diam}\gamma = n^{3/4+o(1)}$ *in probability. The same holds with the length restricted to $n \leq L(\gamma) < n^{1+\delta}$ for any fixed $\delta > 0$. Under the length-weighted law conditioned on diameter $\geq r$ one has $L(\gamma) = r^{4/3+o(1)}$ in probability.*

Finally, the mean length in the unrooted critical ensemble conditioned on diameter $\geq r$ is $r^{4/3+o(1)}$, and the same holds on the power-separated annulus $r \leq \operatorname{diam}\gamma < r^{1+\varepsilon}$.

"In probability" means: for every $\varepsilon > 0$, the value lies between $t^{a-\varepsilon}$ and $t^{a+\varepsilon}$ with probability tending to one as the cutoff $t \to \infty$. The exponents 3/4 (size at fixed length) and 4/3 (length at fixed size) are reciprocal under the relation $D \sim L^{3/4}$, exactly as they should be.

We also prove two boundary (half-plane and slab) results for arches and corridors; those are stated in their own section (§8) because they are logically separate and can be read independently.

1.5 Why this is not obvious, and how the proof is organised

It is tempting to think that the critical mass on big polygons must be a clean power of r because the weight $w(\gamma)$ is a power of a constant per edge. The difficulty is that the sum runs over all shapes, and the leading power of r is not visible from any single polygon. Worse, the natural analytic tool, a complex-valued transfer operator with a family of "charges", produces *complex* expressions whose leading term might cancel. Positivity has to be imported from a separate geometric argument.

The proof goes through a finite cylinder and is organised into five ingredients, each of which is a clean statement. The reader who wants to see only the architecture can look at the five boxed statements below; each is proved later.

(1) A finite polygon identity. Roll the honeycomb lattice onto a cylinder of circumference $N|h|$ (its horizontal period). On the cylinder a maximal self-avoiding polygon is either *contractible* (does not wrap around the tube) or *winding* (wraps). The winding part is what is lost when a planar polygon is projected: a planar polygon wraps if and only if it fails to project injectively. A finite transfer-matrix computation gives an exact algebraic identity (§2, Proposition 2.6)

$$E_N\, \tau_N = c_*\, \tau^v_{N-1}, \qquad c_* = 2 - \sqrt{2},$$

where E_N is the positive winding mass through three sampled edges (one of each honeycomb edge type), τ_N is an unmarked "color sum" attached to the N columns and τ^v_{N-1} is a marked color sum in which one column is distinguished. The left side E_N is positive; the right side is the product of two algebraic quantities. The point of this identity is that it turns a positive geometric mass into a ratio of two algebraic objects whose asymptotics we can compute.

(2) The one-mark ratio. The central analytic estimate (§§5–5.6) is

$$\left|\tau^v_{N-1}/\tau_N\right| = N^{-2/3+o(1)}.$$

This compares the marked and unmarked color sums. The computation runs through an angular (rotated) transfer matrix, then a radius-ordered (radial) series, then a coercivity estimate that controls the complex phases, and finally a spectral model in which an operator on one half-line is joined to an operator on the other half-line by a "marked" link that raises the slope of a conserved lattice momentum by exactly 3/2. That jump $\frac{3}{2}$ becomes the exponent $-\frac{2}{3}$.

(3) The determinant scale. The unmarked color sum has exact size (§4)

$$\tau_N = (\text{base}_1)^{N^2} (\text{base}_2)^N \, N^{-7/12+o(1)}$$

where the two bases are explicit positive constants built from $\sqrt{2}$ and 3. This fixes the denominator in (2).

(4) The two pressures, and their signs. A separate derivative computation on the cylinder (§3) equips the winding mass with two independent positive inputs:

$$A_N := \sum_{\text{winding } \gamma/v} w(\gamma) = c\, N^{-1} + o(N^{-1}) \quad (c > 0), \qquad \sum_{\text{not surviving } \gamma/v} w(\gamma) \sim \tfrac{\gamma_1}{8} N^{-2} \quad (\gamma_1 > 0).$$

The first says the winding mass decays like N^{-1}; the second says the *unrooted* planar mass lost by projection is of order N^{-2}. Both signs are positive because they come from derivatives of a partition function, not from phase cancellation. This is the positivity input the analytic continuation in (2) cannot supply by itself.

(5) Assemble into tails. Combining (1)–(4) gives two projection losses:

$$E_N = \tau^v_{N-1}/\tau_N \cdot c_* \cdot (\text{det factors}) = N^{-2/3+o(1)}, \qquad \sum_{\text{not surviving}} w(\gamma) \asymp N^{-2}.$$

A polygon of diameter $\geq r$ has transverse row span $\gtrsim r$ in one of three symmetric orientations; taking $N \approx r^{1-\varepsilon}$ converts either loss into a planar tail. This gives $A(r) = r^{-2+o(1)}$ and $P(r) = r^{-2/3+o(1)}$ (§6). The second moment $M_2 \leq r^{2/3+o(1)}$ needs the extra two-mark construction of §7, which is where the $N^{5/24}$ Pfaffian normalization and the "intermediate region" appear; its output is the scalar bound

$$|\text{two-mark scalar on the cylinder of width } N, \text{ shorter arc } k| \;\leq\; k^{-4/3+o(1)},$$

and summing over the displacement between the two marked edges gives $M_2 \leq r^{2/3+o(1)}$.

Corollary 1.2 then follows from A, P, M_2 by a purely positive argument (a length bin is split at diameter $n^{3/4}$; mass on the small-diameter half is bounded by M_2, mass on the large-diameter half by A). No further positivity is needed at that last step.

1.6 What the reader needs

Ingredients (1), (4) and the assembly (5) need only finite linear algebra, a transfer matrix on a finite state space, and derivative book-keeping. Ingredient (3) is one contour integral and one reproducing-kernel comparison. Ingredient (2) is the only genuinely analytic part: it invokes a vertex-operator (Frenkel–Kac) construction of a charge lattice, a Heisenberg algebra, a radial-series transformation of the resulting trace, a coercivity estimate on a positive Hilbert space, and a compact-operator spectral calculation. Everything in (2) is standard for a reader who has seen the planar Coulomb gas / vertex operator formalism; for a reader who has not, §5 and §5.3 state each algebraic identity plainly, and §5.6 shows the coercivity and spectral steps in isolation. The full technical discussion of (2) runs from §5 to §5.6.

2 Finite transfer identities

This section builds the finite machine. On a cylinder of circumference $N|h|$, encode self-avoiding polygons in a vector space of noncrossing pairings of the N boundary stubs (the "link states"), with a transfer matrix that slides one row of cells up. A self-avoiding polygon on the cylinder is a linear combination that is a fixed vector of this transfer matrix. We compute those fixed vectors (a polynomial family V_N), specialize them to the physical honeycomb weights, and extract the identity

$$E_N\,\tau_N = c_*\,\tau_{N-1}^{v}$$

that drives the rest of the paper.

2.1 Local weights, braid, fusion and cup

We work with three spin labels: 0 is a vacant edge; $+1$ and -1 are the two orientations of an occupied edge. Take a spectral parameter λ near $d_* = \pi/8$ and put

$$Q_\lambda = e^{i\lambda}, \qquad \chi = i\,e^{-2i\lambda}, \qquad n = \chi^2 + \chi^{-2} = -2\cos(4\lambda), \qquad X_\lambda = \frac{\sin\lambda}{\sin 2\lambda}.$$

At the physical parameter $\lambda = d_*$ write $Q = Q_{d_*}$, $q = Q^2 = \chi = e^{\pi i/4}$, $x_* = X_{d_*} = 1/(Q + Q^{-1}) = (2+\sqrt{2})^{-1/2}$, and $n = 0$.

The local Hilbert space $\mathcal{H} = \mathbb{C}^3$ has basis $\{|0\rangle, |1\rangle, |-1\rangle\}$. On $\mathcal{H}\otimes\mathcal{H}$ we define a matrix $B(r)$, depending on a spectral argument $r = e^{ix}$. It preserves total spin (charge), is transpose-symmetric, is unchanged when both the order of the two tensor factors and the charges are flipped, and satisfies some elementary analytic facts we use throughout: it has poles only at $r^2 = Q_\lambda^{-4}, Q_\lambda^{-6}$ (simple), the values $B(1) = I$, and the two factorizations

$$B(Q_\lambda^3) \propto C\,C^{\mathsf{T}}, \qquad B(Q_\lambda^2) \propto A\,A^{\mathsf{T}},$$

where, with $c = \sum_{a=\pm1} \chi^a |a, -a\rangle$,

$$C = |00\rangle + c, \qquad A|0\rangle = |00\rangle + X_\lambda\,c, \qquad A|a\rangle = X_\lambda(|a,0\rangle + |0,a\rangle) \quad (a = \pm1).$$

The symbols C (a "cup", gluing two legs) and A (a "fusion", merging two legs into one) are the two maps through which all the reductions below go. The entries of B in the diagonal pair (vacuum, both occupied) and the single-occupied block are

$$\begin{pmatrix} t & Uc^t \\ Uc & wI + W c\,c^t \end{pmatrix}, \qquad (t,u,v,U,w,W) = \frac{(p_0 + a_0 b_3,\ s_2 b_3,\ a_0 b_3,\ s_2 a_0,\ b_2 b_3,\ -a_0 b_1)}{g(x)},$$

where $s_j = \sin(j\lambda)$, $a_j = \sin(j\lambda + x)$, $b_j = \sin(j\lambda - x)$, $p_0 = s_2 s_3$, $g(x) = a_2 a_3$, and the two single-occupied eigenvalues are $u + v$ and $u - v$. This is a Yang–Baxter matrix for the dilute $O(1)$ loop model with fugacity n, specialized at $\lambda = d_*$ to the honeycomb SAW weights.

Lemma 2.1 (Braid, unitarity, reductions)**.** *At d_* the following hold as identities of operators on $\mathcal{H}^{\otimes 3}$, with $B_{12} = B\otimes I$ and $B_{23} = I\otimes B$.*

1. ***Unitarity:*** $B(r)B(r^{-1}) = I$.
2. ***Braid relation (Yang–Baxter):*** $B_{12}(z)B_{23}(rz)B_{12}(r) = B_{23}(r)B_{12}(rz)B_{23}(z)$.

3. ***Crossing:*** $g(3\lambda - x)B_{23}(Q_\lambda^3/r)(C \otimes I) = g(x)B_{12}(r)(I \otimes C)$, *and the same with* $12 \leftrightarrow 23$ *and* $C \leftrightarrow C^t$.

4. ***Deleting a pair (cup):*** $B_{12}(r)B_{23}(Q_\lambda^3 r)(C \otimes I) = D(x)(I \otimes C)$, *where* $D(x) = g(-x)/g(3\lambda + x)$ *equals* 1 *at* d_**; similarly with the two factors exchanged.*

5. ***Fusing a pair:*** $B_{12}(r)B_{23}(Q_\lambda^2 r)(A \otimes I) = d_f(x)(I \otimes A)B(Q_\lambda r)$ *with* $d_f(x) = b_3/a_5$*, and the mirror identity; both transport scalars equal* 1 *at* d_*.

Proof. Unitarity follows from $a_j b_j = s_j^2 - a_0^2$ in the single-occupied block plus $w(r)w(r^{-1}) = 1$ in the doubly occupied block. The braid relation is checked at the seven special arguments $z \in \{1, 1/r, Q^3, Q^3/r, 0, \infty, (\text{degenerate})\}$; at each of those B collapses to a degenerate rank and the two sides reduce to the same expression, while generically the entries after clearing $g(x)g(y)$ are polynomials of degree at most eight and parity-fixed, so seven distinct arguments (counted with signs) suffice. Crossing follows by comparing the two blocks in the single- and doubly-occupied entries and matching the χ^{a-b} phases. The cup and fusion identities are the "two special ranks of B" — $B(Q^3) \propto CC^t$ removes a pair, $B(Q^2) \propto AA^t$ replaces a pair by a single spin — applied between the two positions where the pair appears; the transport factors $D(x) = 1$, $d_f(x) = 1$ at d_* because $a_5 = b_3$ and $a_5 = b_3$, $a_6 = b_2$ there. The transport can be verified entry by entry in the single-/double-occupied blocks (see [bookkeeping]: six scalar sine-addition identities $d_f \tilde{t} = tt' + nXUu'$ and its siblings, all elementary). □

Remark 2.2 (Why we want A and C). The matrix B acts on a row of N spins, but the SAW condition — never reuse a corner — is not visible in the spin labels alone. That is why we lift the spins to noncrossing pairings; the two maps C (collapsing a pair to a vacuum, i.e. closing a segment of the loop) and A (replacing a pair by a single occupied leg) are exactly the allowed fusion moves on noncrossing pairings. The braid relation says local sliding of spins past each other is consistent; the reductions say the two collapse moves C, A commute with sliding. The transfer matrix below enumerates noncrossing pairings, and A, C are the tools by which it is reduced column by column.

2.2 Cylinder rows, twists, and the link image

Number the slots $1, \dots, N$ periodically, with a seam between N and 1. Let $z = (z_1, \dots, z_N)$ be site parameters, $t_i = z_i^2$ the squared parameters, and u the auxiliary spectral parameter. The open row with inserted auxiliary spins p on the left seam and k on the right seam is $M_{p,k}(z; u)$; the operator $B(z_i/u)$ propagates through column i. Let $\eta \neq 0$ be a twist and set

$$T_\eta(z; u) = \sum_s \eta^s \, M_{s,s}(z; u).$$

This is a closed transfer matrix on total charge zero on the spatial circle; T_η is rational in $y = u^2$ (parity) and $T_\eta(z; u)$ commutes with itself for different u. The two reductions in Lemma 2.1 yield the product identities

$$T_\eta(z; z_j)T_\eta(z; Q_\lambda^{-2} z_j) = \Big(\prod_i d_f(x_{ij})\Big) T_\eta(z; Q_\lambda^{-1} z_j), \quad T_\eta(z; z_j)T_\eta(z; Q_\lambda^{-3} z_j) = \Big(\prod_i D(x_{ij})\Big) I.$$

At $\lambda = d_*$, crossing gives the transpose symmetry $T_\eta(z; u)^t = T_{\eta^{-1}}(z^{-1}; Q^{-3}/u)$ on total charge zero.

Noncrossing pairings. A *link state* is a noncrossing partial matching of the N spatial slots drawn below the cut of the cylinder, with each arc labelled by the winding class of its two endpoints; unmatched slots are vacant. The empty state (all vacancies) is allowed. This is a finite basis. It maps to charge-zero spin tensors by giving each arc independent spins $s, -s$ ($s = \pm 1$) at its two ends, summing with the factor $\chi^s \eta^{ks}$, where k counts whether the arc crosses the seam. Call the image the *link image*; it is a linear subspace of the spin space, transformed by Ω (rotation by one slot, with twist η) and preserved by projections to any specific set of spin values.

The transfer T_η intertwines the link image with the spinless diagrammatic transfer with loop weights n for a contractible boundary loop and $\eta + \eta^{-1}$ for a winding one. This is the snake / Temperley–Lieb re-indexing: insertions of B locally act on two spatial slots and preserve the link image because B's local blocks are the six local connection weights t, u, v, U, w, W. At $\lambda = d_*$ and $\eta = \pm i$ two facts hold: (i) the empty-cell weight $t = 1$ identically, so the vacuum of the link-image diagram is uniform; (ii) all closed loops have $n = 0$ weight at $\eta = \pm i$ under the physical pull-back, so the sum over closed contractible polygons is killed and only winding ones survive at first order.

The two physical reductions. A pair $t, q^2 t$ of column parameters of type A fuses to the single column qt; a pair $t, q^3 t$ of type C deletes. These are the physical meaning of Lemma 2.1: $B(Q^2) = AA^t$ fuses two columns of the polynomial vector to one, $B(Q^3) = CC^t$ removes two columns. At the homogeneous point $z_i = 1$ with auxiliary $u = Q^{-1}$ these fusions are exactly the two grouping choices on each honeycomb cell: the bottom-left and top-right ports group together (one honeycomb trivalent vertex), the bottom-right and top-left ports group (the other). The line between the bottom and top row is a horizontal cut through the edge midpoints.

2.3 The two polynomial vacua

The fixed vectors of T_η have a rational structure in the site parameters that we now exploit. We work at $\lambda = d_*$ throughout and use two twists:

$$p = 2, \quad \eta = i = q^2; \qquad\qquad p = 1, \quad \eta = -i = q^{-2}.$$

The two are related by the particle-hole exchange, and both are needed because the polygon mass Z_η (next subsection) has different signs on contractible and winding polygons for $\eta = i$ vs. $\eta = -i$.

Proposition 2.3 (The two polynomial vacua)**.** *For each choice $(p, \eta) \in \{(1, -i), (2, i)\}$ there is a unique family of polynomial vectors $V_N(z)$, $N \geq 0$, in the link image, with $V_0 = 1$, $V_1 = |0\rangle$, satisfying:*

- homogeneity: *the z-degree is $pN(N-1)$, and after stripping the spin-s_l component by $\prod_l z_l^{|s_l|}$, each coefficient is polynomial in the $t_l = z_l^2$ of degree at most $p(N-1) - |s_l|$ in t_l;*
- exchange/rotation: *exchanging two adjacent arguments acts by $B_{ij}(z_i/z_j)$, and a left rotation acts by Ω on the charges;*
- deletion: *if the spin at slot l is 0, the vacuum evaluates at $t_l = 0$ and $t_l = \infty$ to V_{N-1} with slot l deleted (up to the appropriate powers of t_k);*
- pair fusions: *at $z_j = Q^3 z_i$ the vector takes the value prescribed by inserting a cup C on V_{N-2} of the other slots and multiplying by an explicit prefactor Π_{ij} (see below); for $p = 2$ this holds for both orders $i \to j, j \to i$, for $p = 1$ for $i < j$.*

The empty coefficient $D_p = V_N^{empty}$ is not identically zero, and

$$T_\eta(z;u)V_N(z) = V_N(z)$$

as a rational identity in u and z. The normalized vector V_N/D_p extends regularly to homogeneous site parameters and equals the empty-normalized stationary state obtained from the physical transfer-semigroup limit $\lim_{m\to\infty} T_\eta^m e_\varnothing$.

Proof sketch. The proof is an induction on N. The base cases $N = 0, 1, 2$ are displayed. For $N \geq 3$ one interpolates in the last site parameter t_N: prescribe the value of V_N on the pair slices $t_N = q^{\pm 3} t_i$ (for $p = 2$, both signs; for $p = 1$, positive only), plus the deletion value at $t_N = 0$, and Lagrange-interpolate. The prescribed inputs W_{ij} are polynomial on the pair slices (Lemma 2.1 kills the exchange poles there), so the interpolation has only removable poles and yields a polynomial with the correct degree. Compatibility of two intersecting pair slices, and covariance of the prescriptions under the exchanges and rotations, come from the braid relation and the reduction identities $B(Q^3) = CC^t$, $B(Q^2) = AA^t$ on the pair slices. This closes the induction for the pair values, exchanges, and rotation. The identity $T_\eta V_N = V_N$ is proved by comparing both sides as polynomials in one site variable t_l: the difference vanishes at the pair nodes by the Π_{ij} prescriptions, at $t_l = u^2$ and $t_l = q^3u^2$ by the two special ranks of B, and at 0 by the deletion limit; a degree count then kills the difference. Uniqueness follows from the same degree count at every induction step. Regularity at the homogeneous point follows from the interpolation representation together with the physical transfer-semigroup argument of §2.2: the physical transfer is nonnegative, contracts coefficientwise to a rank-one eigenvector, and this normalized eigenvector agrees with V_N/D_p by generic uniqueness. □

The empty component is a Pfaffian. For the $p = 1$ vacuum (the $-i$ twist) with N even, the empty coefficient is an alternant of 2×2 minors of a single Cauchy-like kernel:

$$D_1(X) = \prod_{i<j} \frac{p_K(x_i, x_j)}{x_j - x_i} \operatorname{Pf}\left[\frac{x_j^2 - x_i^2}{p_K(x_i, x_j)}\right]_{i,j=1}^{N}, \qquad p_K(x, y) = x^2 + \sqrt{2}\, xy + y^2. \tag{4}$$

The prefactor clears the apparent poles of p_K in the rows and columns of the Pfaffian, so this is a symmetric polynomial of the required degree. Its two crucial values on pair slices are

$$D_1(x_1, x_2) = x_1 + x_2, \qquad D_1(Y, q^3 x) = \sqrt{2}\, x \prod_{t\in Y} (t - x)\, D_1(Y) \text{ (the } q^3 \text{ deletion)},$$

and the $q^{\pm 1}$ fusion identity

$$D_1(Y, q^{-1}x, qx) = \sqrt{2}\, x \prod_{t\in Y} (t + x)\, D_1(Y, x).$$

For the $p = 2$ vacuum (the $+i$ twist) the empty component is the squared Vandermonde form

$$D_2(X) = P^N \prod_{i<j} \left[\frac{p_K(t_i, t_j)}{t_i - t_j}\right]^2 \det[k(t_i, t_j)], \qquad k(t_i, t_j) = \frac{t_i t_j}{p_K(t_i, t_j)}, \tag{5}$$

with $P = 2 + \sqrt{2}$. Both expressions are nonzero; in particular the transfer has a unique physical stationary state and both continuations are well defined.

2.4 A marked row and its polygon interpretation

A closed transfer kills every closed loop when $\eta = \pm i$: that is why the unmarked stationary state is a bare vacuum. To measure the polygon mass that sits in one particular edge, freeze the orientation of a single auxiliary edge instead of summing it. For an operator H on the N-site spin space write the bilinear bracket

$$\langle H \rangle = \Psi_\eta(z^{-1})^t \, H \, \Psi_\eta(z), \qquad Z_\eta(y) = \langle M_{-,-}(z; \sqrt{y}) \rangle,$$

where $\Psi_\eta = V_N / D_p$ is the normalized vacuum and $M_{p,k}$ is the open row with fixed auxiliaries p on the left seam and k on the right seam. Write $P_{l,s}$ for the projector to spin s at slot l.

Proposition 2.4 (Marked-edge count)**.** *At homogeneous site parameters and $y = q^{-k}$, $k = 1, 2$, the value $Z_\eta(q^{-k})$ is a sum over self-avoiding polygons through the sampled edge, with weight $x_*^{|\gamma|}$, of signs $+1$ or -1 depending on the parity of the loop. Precisely, at the minus twist $\eta = -i$ the three values $Z_{-i}(1), Z_{-i}(q^{-1}), Z_{-i}(q^{-2})$ each equal x_*^2 times the sum over* positive *(both essential and contractible) polygon mass through one fixed edge; the three edges $1, q^{-1}, q^{-2}$ sample one of each honeycomb edge type, up to lattice translations on the same cylinder. At the plus twist $\eta = +i$ the same sum is essential mass* minus *contractible mass. The values pass to the limits because the minus twist makes the sum positive and bounded.*

Proof. The projection to the fixed auxiliary edge closes one polygon at that edge. All remaining closed loops cancel in pairs by the local turn phase: a contractible simple loop contributes ± 1 with squared total zero, a winding one contributes $\pm i$ with squared zero; the surviving loop is the unique one through the mark, because its orientation is uniquely fixed by the mark. This residual sign is $+1$ at $\eta = -i$ and ± 1 depending on whether the surviving loop is contractible or essential at $\eta = +i$, as computed by the seam-crossing count N_L, N_U and $q^{\pm 2}\eta^{\pm 1}$. Summing over all three edge types gives positive mass at the minus twist; the sum over essential loops alone at $+i$ follows by subtracting. □

The important point is that at $\eta = -i$ the marked scalar Z_{-i} is a positive geometric quantity, whereas at $\eta = +i$ it is the difference of essential and contractible parts. Since the polymer mass is positive, the combination

$$S_a := c_* + Z_{+i}(a) + Z_{+i}(q^{-1}a) + Z_{+i}(q^{-2}a) = Z_{-i}(a) + Z_{-i}(q^{-1}a) + Z_{-i}(q^{-2}a) - c_*$$

is real and its size gives the winding contribution; here $c_* = 2 - \sqrt{2} = 2/P$ and a is a distinguished site parameter.

2.5 Finite identities for marked polygon masses

We now write the marked scalar in terms of finite "color sums," which are the algebraic objects whose asymptotics we will compute.

Let $d = q + q^{-1} = \sqrt{2}$, $P = 2 + d$, $c_* = 2 - d = 2/P$. For a site list $X = (t_i)$ with a distinguished parameter $a \in X$, define the summing object

$$S'_{\eta,a} = Z_\eta(a) + Z_\eta(q^{-1}a) + Z_\eta(q^{-2}a), \qquad S_a = c_* + S'_{i,a},$$

where in the subscript $i = \sqrt{-1}$. Note the three values sit at three distinct arguments.

The unmarked color sum. Introduce three colors $\{-,0,+\}$ with weights

$$\ell_- = \frac{-1-d+i}{2}, \quad \ell_0 = P, \quad \ell_+ = q\ell_-, \qquad \ell_- + \ell_0 + \ell_+ = 1, \quad \ell_-\ell_+ = P^2/\beta, \quad \beta = G(1) = 2P,$$

and a pair weight between two sites of colors s,t, in a fixed order $(x,y) \mapsto$ factor $g_{st}(y/x)$ given by the symmetric table

$$g(r) = \begin{pmatrix} 1 & G(r/q) & G(r/q)G(r/q^2) \\ G(rq) & G(r) & G(r/q) \\ G(rq)G(rq^2) & G(rq) & 1 \end{pmatrix}, \qquad G(t) = \frac{(1+t)^2}{(t-q)(t-q^{-1})}.$$

The symmetry $g_{st}(r) = g_{ts}(1/r)$ makes the pair product reordering-invariant. Sum over all 3^N color assignments; the result is the *unmarked color sum* τ_X. Also let $F_X(y)$ be the same sum with one site decorated by an extra explicit rational factor $L_\sigma(t/y)$, where

$$L_- = L = \frac{f_5 f_6 f_7}{f_2 f_3 f_4}, \quad L_0(r) = \frac{L(r/q)}{L(r/q^2)}, \quad L_+(r) = \frac{1}{L(r/q^3)}, \qquad f_j(t) = 1 - q^j t.$$

Define

$$R_X = \frac{D_+(X)^2}{\prod_{\{x,y\}\subset X} \delta(x,y)}, \qquad \delta(x,y) = p_K(x,y)(x^2+y^2),$$

where $D_+ = D_2$ is the Pfaffian of (5). Both τ_X and R_X are symmetric rational functions of the t's. The central identity is that they coincide.

Proposition 2.5 (Determinant equals color sum)**.** *As rational functions, $R_X = \tau_X$.*

Proof. Both sides are rational, so it suffices to check that their poles agree (residue by residue) and that both have the same limits at $t_i = 0, \infty$. The only possible poles of τ_X are at $t_j = q^{\pm 2} t_i$ (fusion of two colors) and $t_j = q^{\pm 3} t_i$ (deletion), each simple and computed by a short table evaluation: two colors combine to a fused color, with weight αP at the q^2 fusion and $\alpha G(q^2)P^2/\beta$ at the q^3 deletion. The determinant R_X has the same simple poles at the same arguments (the identity $k_Y^\perp(s,t) = H(s/a)H(t/a)k_{Y'}^\perp(s,t)$ for the Pfaffian Schur complement, with $H(t) = (t^4-1)/(t^4+1)$, shows the squared determinant transforms by $\prod G$ exactly as the color pair table), with the same residues. Both sides have the same $\prod_{x<y} t_x^{-2(N-1)}$ behaviour under $t_x \mapsto 1/t_x$ (the inversion identity), and both tend to det-level or colored-vacuum constants at $t \to 0, \infty$. The induction is closed at the initial values $N = 0, 1, 2$. □

Decorating the distinguished site. The three values $F_X(a), F_X(q^{-1}a), F_X(q^{-2}a)$ coincide; denote the common value $F_a(X)$. Each equals one half of the sum over the other sites with the additional explicit pair factor H_s^v displayed in [bookkeeping]. Concretely, $2F_a(X) = \tau^v$ where τ^v is the color sum on the $N-1$ *other* sites with one distinguished site's worth of decoration. The central algebraic identity is

$$S_a\, R_X = 4c_*\, F_a. \tag{6}$$

Proof of (6) *(sketch).* Both sides are rational and symmetric in the generic spectra; it suffices to identify them via their residues at the special arguments $t_j/a = q^{\pm 1, \pm 2, \pm 3}$ and at -1. The pair-reduction identities from Lemma 2.1 (fusion q^2 and deletion q^3) give the matching at $q^{\pm 2}, q^{\pm 3}$; the cancellation at $q^{\pm 1}$ comes from the two distinct sums in S_a. All this leaves a small polynomial ambiguity in the residues of b/a at $\pm 3, -1$; symmetry reduces it to at most one unknown constant. The cases $N = 1, 2, 3$ are computed explicitly and pin the unknown constant. This is the "three initial evaluations" carried out in detail in the original. □

The finite polygon identity. Let τ_N be the value of τ_X at all N homogeneous sites, and τ^v_{N-1} the value of the decorated sum on $N-1$ homogeneous sites.

Proposition 2.6 (Finite polygon identity)**.** *For the honeycomb cylinder of horizontal period Nh and even N, let E_N be the sum of $x_*^{|\gamma|}$ over* winding *self-avoiding polygons through three sampled edges, one of each honeycomb edge type, counted once for each sampled edge used. Then*

$$E_N\,\tau_N = c_*\,\tau^v_{N-1}, \qquad c_* = 2-\sqrt{2}. \tag{7}$$

Moreover, the total mass of all *self-avoiding polygons (winding plus contractible) through the three sampled edges is exactly c_*.*

Proof. By Proposition 2.4, $S'_{-i,a}$ counts the sum of winding and contractible polygons, $S'_{+i,a}$ their difference, so that $S_a = c_* + S'_{+i,a} = S'_{-i,a} + S'_{+i,a} = 2E_N$. Substitute $S_a = 2E_N$, $R_X = \tau_N$ (Proposition 2.5) and $F_a = \tau^v_{N-1}/2$ into the marked-scalar identity (6). Dividing by 2 gives (7). The all-polygon total is $S'_{-i,a} = c_*$, obtained by a parallel evaluation of (6) with the $+i$ and $-i$ decorations combined. □

This is ingredient (1): a positive winding mass expressed as a ratio $E_N = c_*\tau^v_{N-1}/\tau_N$ of two finite color sums. The denominator is nonzero (Proposition 2.3), and §5–§5.6 evaluates the ratio. It remains only to extract asymptotics.

3 Two pressures on the cylinder

The finite identity (7) expresses E_N as a ratio of τ^v_{N-1} and τ_N. To know the ratio is real and positive we need an independent positivity input, because tracking τ^v and τ_N separately through the analytic continuation of §5–§5.6 lets complex phases cancel. This section supplies two such inputs as derivatives of a positive partition function.

Fix a cylinder of circumference $N|h|$. The closed transfer on the link image carries two loop weights: a factor $n = -2\cos(4\lambda)$ for each contractible closed loop and a factor $s = \eta + \eta^{-1}$ for each winding one (the book-keeping is §2.2). At $\lambda = d_*$, $n = 0$ and $s = 0$ at $\eta = \pm i$; we work near $\eta = i$ and use s as a small parameter.

Let $\Lambda(y)$ be the eigenvalue of the cylinder transfer on the link image, chosen continuously near 1 at the physical point; the notation is $y = u^2$ for the squared auxiliary parameter. As a function of the twist s it has a Taylor expansion at $s = 0$; define the *responses*

$$J_m(y) = [s^m]\Lambda(y)\Big|_{\lambda=d_*}, \quad m \ge 1; \qquad J(y) = \partial_\lambda \Lambda(y)\Big|_{\lambda=d_*,\ s=0}.$$

The first class is a twist-derivative (moving the winding loop weight), the second is a λ-derivative (moving the local cell weights while keeping the twist zero).

Lemma 3.1 (Physical evaluations)**.** *At homogeneous site parameters $z_i = 1$ and $y = q^{-1}$,*

$$J_1(q^{-1}) = A_N, \qquad J(q^{-1}) = N(\log t_{\mathrm{phys}})'(d_*) + 8A^0_N,$$

where

$$A_N = \sum_{\text{winding } \gamma/v} x_*^{|\gamma|}, \qquad A^0_N = \sum_{\text{contractible } \gamma/v} x_*^{|\gamma|},$$

are the winding and contractible masses (sums over axial translation classes, "γ/v"), and $t_{\mathrm{phys}}(\lambda)$ is the empty-cell weight at $u = Q_\lambda^{-1}$.

Proof. On a finite-height cylinder the partition function is exactly a gas of vertex-disjoint polygons with step weight X_λ, loop weights n, s and empty scalar t_{phys} per cell. Its logarithm per row converges to the leading log-eigenvalue by the fixed-N spectral gap: the vacuum matrix element of the leading projection is analytic and nonzero nearby. At $s = 0$ (physical twist) the factor $\Lambda = 1$ identically in y, so the first derivative of $\log \Lambda$ is the first derivative of Λ. Differentiating in s with $\lambda = d_*$, $n = 0$ selects exactly one polygon (the ∂_s pulls down one winding loop). Differentiating in λ at $s = 0$ splits into the empty-cell scalar $N(\log t_{\mathrm{phys}})'(d_*)$ and a one-polygon term with weight $n'(d_*) = 8$, i.e. $8A_N^0$. Both are monotone in the height cut-off (extra height adds positive mass); the limits exist by monotone convergence. □

So the two responses compute the two sums we care about. The remainder of this section shows that both leading constants are of definite sign, by writing each response as the convolution of a positive boundary function with an explicit kernel.

3.1 Fusion constraints and the convolution operator

At $y = \infty$ and $y = 0$,

$$\Lambda(y) = (1 + s)(-e^{\pm 8i\lambda})^N,$$

from the two extreme weighted swaps (other permutations of the auxiliary cannot close). The two reductions of §2.2 give

$$\Lambda(t_j)\Lambda(q_\lambda^{-2}t_j) = \Big(\prod_i d_f(x_{ij})\Big)\Lambda(q_\lambda^{-1}t_j), \qquad \Lambda(t_j)\Lambda(q_\lambda^{-3}t_j) = \prod_i D(x_{ij}),$$

with $d_f, D = 1$ at d_*. Consequences:

- For J_m (twist response), $\Lambda(qy)\Lambda(q^{-1}y) - \Lambda(y)$ has order-N zeros at $q^{-1}t_j$ and $q^{-2}t_j$.
- For J (contractible response), the same holds for $\Lambda(qy)\Lambda(q^{-1}y) - \Lambda(y) - S_X(y)$, where

$$S_X(y) = 8i \sum_l \frac{y - t_l}{y + t_l}.$$

Next, transform to a variable in which the constraints become an accretive convolution. Put

$$y = e^{-3id}e^w, \qquad h_0 = 3d, \qquad \alpha = \pi/(6d) = 4/3,$$

and write each response as a rational function of e^w; it has poles only at $w = \pm id + i\pi$ (order at most N). Partial fractions expand each $V(w)$ (a J_m or J) as a constant plus a finite combination of $\partial_w^k H_0(w \pm id)$ for $0 \le k < N$, $H_0(w) = e^w/(1 + e^w)$. Define $v(w)$ by replacing $H_0(w \pm id)$ by $H_0(\alpha(w \pm id))$ under the same derivatives.

Then on horizontal lines $|\Im w| \le h_0$, one has

$$V(w+2id)+V(w-2id)-V(w) = \mathsf{T} * v, \qquad \widehat{\mathsf{T}}(k) = \alpha(2\cosh(2dk)-1)\frac{\sinh(6dk)}{\sinh(8dk)}, \quad \widehat{\mathsf{T}}(0) = 1, \tag{8}$$

and the symbol satisfies $\widehat{\mathsf{T}}(k)/\alpha = 1-$ (small positive correction), so T is a real even measure: an atom of mass α at 0 plus a positive exponentially decaying density. In the two norms

$$\|b\|_\pm^2 = \sum_{\phi=\pm h_0} \int_{\mathbb{R}} |b(x + i\phi)|^2 \alpha e^{\pm\alpha x}\, dx$$

(it is the sum over the two horizontal lines; the sign $+/-$ of the summing weight is independent of the line) the operator T is bounded and strictly accretive: $\Re\langle b, \mathsf{T} * b\rangle_\pm \geq c_\mathsf{T}\|b\|_\pm^2$ for some $c_\mathsf{T} > 0$. This follows from Plancherel after weighting by $e^{\pm\alpha x}$: on the shifted line $k \pm i\alpha/2$ the fraction subtracted in $\widehat{\mathsf{T}}/\alpha$ has modulus uniformly less than 1.

Two further convolution facts. First $\mathsf{T} = \mathsf{T}_+ * \mathsf{T}_-$ with real measures supported on $[0,\infty)$ and $(-\infty, 0]$, whose inverses are supported the same way and which have exponential moments at rate $> \alpha/2$ in total variation. Second, if $F = V(\cdot + 2id) + V(\cdot - 2id) - V$ then $V = L * F$ with

$$\widehat{L}(k) = (2\cosh(2dk) - 1)^{-1}, \qquad L(\pm x + ip) = c_L e^{-\alpha x} e^{\mp i\alpha p} + O_\epsilon(e^{-2\alpha x})\ (x \to \infty), \quad c_L = \frac{1}{2\sqrt{3}d},$$

uniformly for $|p| \leq 2d - \epsilon$. The kernel $L(x + ip)$ has exponential decay $e^{-\alpha|x|}$ with a simple exponential leading term.

3.2 The essential response: $A_N \sim c/N$

Here $V = J_m$, so $F_m = \mathsf{T} * f_m$ with $f_m = \mathsf{T}^{-1}(V)$, and we focus on $m = 1$. The limits of J_m, f_m, F_m at the ends $w = \pm i\infty$ are all $\delta_{m=1}$.

Proposition 3.2 (Essential twist response)**.** *The limit*

$$\lim_{N\to\infty} N\, J_1(id) = 2c_L\alpha^{-1}\cos(\alpha d)\int_0^\infty F_{\text{edge}}(u)\, du > 0,$$

where $F_{\text{edge}}(u) > 0$ is the boundary function

$$F_{\text{edge}}(u) = \lim_{N\to\infty} u\, b_N(u)^{-1} F_1(x + id), \qquad b_N(u) = B_1(u/N)^N,$$

$$B_1(z) = \frac{(z-a)(z-\bar{a})}{(z+a)(z+\bar{a})}, \quad a = e^{i\pi/6}.$$

In particular $A_N \sim c/N$ with $c > 0$.

Proof sketch. B_1 is inner on the right half-plane and invariant under $z \mapsto 1/z$; its boundary of coefficient growth is the two "edges" $u_e = N/z$ (right) and $u_e = Nz$ (left). On each edge $b_N(u_e) \to b_\infty(u_e) = e^{-bu_e}$, $b = 2\sqrt{3}$. Orthogonality of $f_m - c_m B_1$ against $G_m - d_m B_1$ in the $+/-$ norms together with accretivity gives the norm bound

$$\|f_m - c_m\|_\pm + \|G_m/B_N - d_m\|_\pm \leq C_m\sqrt{N},$$

and hence a pointwise bound on J_m off the poles, of the shape $\min(1, e^{\alpha|x|}/N)$ on $|\phi| \leq 5d - \epsilon$. From these, taking a weak-L^2 limit of the edge-scaled functions $g_N = (f_1 - b_N)/u_e$ and $h_N = (F_1/b_N - 1)/u_e$ along the two edges, one shows that g_N has nonnegative Fourier density on $u_e > 0$ and that the density of g is supported in $[0, b]$ in the $x' = \log(t/b)/\alpha$ coordinate, with $\varphi(t) = \mathbf{1}_{t>b}$ (boundary value). The convolution identity passes to the limit and factors uniquely:

$$q_{\text{edge}} = \mathsf{T}_+^{-1} * \psi = \mathsf{T}_- * \varphi = c'\mathbf{1}_{(0,\infty)}, \qquad c' > 0,$$

which pins F_{edge}. Finally $J_1(id) = \int L(id - x)F_1(x)\, dx$; changing variables to $u_e = Ne^{-\alpha|x|}$ and dominating gives the displayed limit as $N \to \infty$, and the integral is positive. (Continuity and compactness of the convolution measure make all weak limits strong and sequence-independent.) $\square$

3.3 The contractible response: $J(id) = N\gamma_0 - \gamma_1/N + o(1/N)$

Now $V = J$, the derivative in λ. Its right and left edge limits are $8iN$ and $-8iN$, of order N (the empty-cell term dominates). Write $S(w) = 8iN(2H_0(w - 3id) - 1)$, $G = F - S$, and $g = \mathsf{T}^{-1} * S$. Then G has the order-N fusion zeros; the forced term g is analytic in a slightly wider strip than $|\Im w| \le h_0$, and the closest poles of $\widehat{g'}(k)$ are double at $\pm i\alpha$, with coefficients $\pm NA_*$, $A_* = 4\pi/(3\sqrt{3}d^2) > 0$. Shifting contours past these poles gives

$$g - j_e = u_e\left[\kappa(\log N - \log u_e) + k_e + O((|u_e|/N)^\delta)\right], \quad \kappa = A_*/\alpha^2 > 0,$$

on $|u_e| \le N$, with $\delta > 0$; j_e is the edge limit of J.

Proposition 3.3 (Contractible pressure)**.** *The contractible response satisfies*

$$J(id) = N\gamma_0 - \gamma_1/N + o(1/N), \qquad \gamma_1 > 0,$$

with $N\gamma_0$ the empty-cell term.

Proof sketch. This is the same machinery as the essential response. The crucial change is that the boundary density of $e_N = (f - g)/u_e$ acquires a logarithmic profile: the projection identity $P_+(e_N/b_N) = -P_+Q_N$, with $Q_N = (g - j_e)/(u_e b_N)$, together with the log-jump of P_+Q_N at the two banks of the negative real axis, forces $\varphi(t) = -\kappa/t$ for $t > b$. Then $\psi = \mathsf{T} * \varphi$ on $x' > 0$ has $q_{\text{edge}} = -(\kappa/b)c_- e^{-\alpha x'}$ with $c_- > 0$, and applying T_+ gives

$$\int_0^\infty G_{\text{edge}}(u_e)\, du_e = -\frac{\kappa c_-}{2b^2}\int e^{-\alpha s'}\mathsf{T}_+(ds') < 0.$$

Restoring the boundary term $N\gamma_0$ (the empty-cell contribution) using $|G(x)| \le C\sqrt{u_e}\exp(-cu_e)$ on $u_e = Ne^{-\alpha|x|}$ gives precisely the stated form with $\gamma_1 > 0$. □

3.4 Putting the two responses together

The physical evaluations of §3.1 together with Propositions 3.2 and 3.3 give

$$A_N = J_1(id) \sim c/N \quad (c > 0), \qquad J(id) = N(\log t_{\text{phys}})'(d_*) + 8A_N^0, \tag{9}$$

so

$$A_N^0/N = (\gamma_0 - (\log t_{\text{phys}})'(d_*))/8 + \frac{\gamma_1}{8N^2} + o(N^{-2}).$$

The first term is the total unrooted plane activity summed over survivors of the projection to period Nh; each surviving plane polygon gives N axial classes (bijection by lifting the projection). Therefore

$$\sum_{\text{not surviving } \gamma/v} w(\gamma) = \frac{\gamma_1}{8N^2} + o(N^{-2}). \tag{10}$$

This is the two-sided version of the unrooted projection loss: mass N^{-2}, coefficient $\gamma_1/8$ positive.

3.5 Thin cylinders suppress large transverse spans

The last ingredient of the pressure section is the statement that cylinder polygons of tall transverse row-span are negligible at any polynomial order.

Proposition 3.4 (Thin cylinders)**.** *Let $H(\gamma)$ be the row span of a cylinder polygon of period Nh. For every fixed $\delta > 0$, $p \geq 0$ and integer $K > 0$,*

$$\sum_{\gamma/v:\ H(\gamma)\geq N^{1+\delta}} (1+|\gamma|)^p\, w(\gamma) = O(N^{-K}).$$

Proof. Connected tuples. The m-th log coefficient in s of the winding partition function is $1/m!$ times a weighted sum over ordered m-tuples with a factor $\sum(-1)^{\#\text{edges}}$ over connected spanning subgraphs of their incompatibility graph. For a connected graph this factor has sign $(-1)^{m-1}$ and magnitude at least 1 (induction by deletion/contraction of a single edge); for disconnected it is zero. Passage to axial classes and (9) bound the total connected-tuple weights by C_m/N.

Winding spans. For an absolute winding polygon γ let M_γ be the total weight of all absolute winding polygons sharing a vertex with it, including itself. If $H(\gamma) > 2$, a subpath between two same-sublattice vertices has displacement $D_\gamma = i'h + j'v$ with $|j'| \geq H(\gamma) - 2$. Periodize by D_γ: every coset of the lattice modulo Nh, D_γ meets γ at some translate of any other winding type (a fine-winding argument on a large circle), giving at least $(H(\gamma) - 2)A_N$ neighbors in the full translation orbit. Restricting the tuple bound to stars gives

$$\sum_{\text{ess}/v} w(\gamma)(1+H(\gamma))^{m-1} \leq C'_m N^{m-2}.$$

Since $|\gamma| \leq 2N(H+1)$, this proves the claim for windings.

Contractible spans. For a tall contractible polygon δ', lock an $R_{i,j}$ in its highest row. Windings with lowest row $j+1$ occupying $P_{i,j+1}$ weigh at least A_N/N. Take the symmetric edge difference with the hexagonal face spanned by the two rows; the overlap with the face is a single short path in each polygon, so this splices into a single simple winding polygon ζ with $H(\zeta) \geq H(\delta') - 1$ and length within 2. Moreover at most $C|\zeta|$ inverse joins per output class, giving a bound $A_N/N \cdot$ (tall contractible) by a relaxed tall essential sum with a linear weight, which is already negligible. This proves the claim for contractibles. □

4 The unmarked determinant scale

We must know the size of the denominator τ_N in the finite identity (7). This is the value of the unmarked color sum τ_X at the homogeneous point. Proposition 2.5 writes it as $R_X = \tau_X$, and at the homogeneous point the sum R_X is the square of a Pfaffian $D_+(X)$ (the empty component of the $p = 2$ vacuum), divided by a product $\prod_{i<j} \delta(t_i, t_j)$:

Proposition 4.1 (Determinant scale)**.** *With a and b the two explicit positive constants*

$$a = P^2 = \left(2+\sqrt{2}\right)^2, \qquad b = \text{an explicit constant built from } \sqrt{2} \text{ and } 3,$$

one has, for even N,

$$\tau_N = a^{N^2}\, b^N\, N^{-7/12+o(1)}, \qquad N \to \infty. \tag{11}$$

In particular $\tau_N \neq 0$, and the ratio τ^v_{N-1}/τ_N in the finite identity has size $|\tau^v_{N-1}|/\tau_N$ exactly (no cancellation).

Proof. We give the argument in five steps.

Step 1: a positive Fourier kernel. Write the Pfaffian kernel $k(t,s) = ts/p_K(t,s)$ with $p_K(t,s) = t^2 + \sqrt{2}\,ts + s^2$. In log-coordinates $w = \log(t/s)$ this equals $\sinh w/(\cosh w + 1/\sqrt{2})$, and a residue calculation on the two branches $|t| = 1$ and $|s| = 1$ of the Cauchy kernel gives the Fourier expansion

$$k(t,s) = \frac{2ts}{t^2+s^2}\int_0^\infty \rho(2T)\,\cos\big(T\log(t/s)\big)\,dT, \qquad \rho(k) = \frac{k\cosh(\pi k/4)}{\sinh(\pi k)}. \tag{12}$$

The weight ρ is positive, even, and decreases like $ke^{-\pi k/4}$ at $k \to +\infty$; near $k = 0$ it has a simple pole $\rho(k) \sim 1/\pi$. The pole at zero is harmless once the short-distance structure of $k(t,s)$ has factored out, and the Fourier transform of ρ is a strictly positive Schur-test kernel. The same must be done for both $k(t,s)$ and $k(t,t)$-level self-interactions; it gives the *ratio* of the homogeneous Pfaffian $\widetilde{D}$ to its self-factor as an expectation under a probability measure supported on $T_k > 0$.

Step 2: comparison Pfaffian. Let σ_1 be a second positive even weight, $\sigma_1(k) = k/\sinh(\pi A_0 k)$ with $A_0 = 3/4$, and $\sigma_{1/2}(k)$ the "flat" variant proportional to $1/\cosh(\pi A_0 k)$. The three weights $\rho, \sigma_1, \sigma_{1/2}$ have pairwise monotonically-ordered ratios:

$$\frac{\rho}{\sigma_1} \text{ is decreasing}, \qquad \frac{\rho}{\sigma_{1/2}} \text{ is increasing}, \qquad \frac{\sigma_1}{\sigma_{1/2}} \text{ is monotone the other way},$$

each ratio proportional (uniformly in $k > 0$ with $u = \pi|k|/4$) to an explicit elementary function of u; the versions are the extremal evaluations of the same signed measure problem.

Step 3: Pfaffian ratio. Order the angles of the homogeneous site parameters. Expanding the comparison Pfaffian by wedging its two-vector integral kernels gives a probability law on $N/2$ variables $T_k > 0$ with density

$$\det[e^{\pm\theta_j T_k}]_{j;(k,\pm)} \prod_k \frac{dT_k}{\sinh(\pi b' T_k)}.$$

The sign is constant (total positivity of the exponentials; alternation within each pair makes $T_k = 0$ integrable and the angular width gives $T \to \infty$ integrability; the common phases cancel between the two Pfaffians). Therefore the ratio of the two Pfaffians is a strictly positive expection,

$$\mathcal{R}_N = \frac{\text{original kernel Pfaffian}}{\text{comparison kernel Pfaffian}} = \mathbb{E}\prod_{k=1}^{N/2} r(T_k) > 0, \qquad r(T) = \frac{(1+e^{-u})(1-e^{-3u})}{1-e^{-4u}} = 1+\frac{1}{2\cosh u} \geq 1, \quad u = \frac{\pi T}{2}$$

Jensen's inequality gives

$$\log \mathcal{R}_N \geq \int_0^\infty \rho_N(T)\log r(T)\,dT, \tag{13}$$

where $\rho_N(T) = \mathbb{E}\sum \delta(T - T_k)$ is the empirical density of T's. Since $\log r \geq 0$, $\mathcal{R}_N \geq 1$; but we need the actual size, which comes from the density ρ_N.

Step 4: Meixner-Pollaczek asymptotics. The comparison polynomials are the Meixner-Pollaczek family $P_d^{(\lambda)}(3k/4;\pi/2)$ in their standard generating-function normalization. Under the measures $\sigma_1, \sigma_{1/2}, \rho$, the moment problem is classical, and the corresponding moment determinants are the Pfaffians of the respective weights. The reproducing kernel of degree $< N$ polynomials under $\sigma_0 r^t$, $0 \leq t \leq 1$, satisfies

$$\sigma_0(T)\,r(T)^t\,\mathcal{K}_{N,t}(T,T)/\log N \to b'/\pi,$$

dominated by a polynomial in $|T|$, uniformly in T compact. With $\alpha = 1/2 + ib'T$ on just this display, the Fourier generating function factors as $(1+iz)^{-\alpha}(1-iz)^{-\bar{\alpha}}$; the corresponding orthogonal polynomials have the classical asymptotic

$$q_n(T) = n^{-1/2}\big[(-i)^n n^{ib'T}\mathcal{A}(T) + i^n n^{-ib'T}\overline{\mathcal{A}(T)}\big] + o(n^{-1/2}),$$

$\mathcal{A}(T) = 2^{-\bar{\alpha}}/\Gamma(\alpha)$, and $|\Gamma(\alpha)|^2 = \pi/\cosh(\pi b' T)$. From this,

$$\mathcal{K}_{N,0}(T, T + v'/\log N)/\log N \to 2|\mathcal{A}(T)|^2 \frac{\sin(b'v')}{b'v'}.$$

The one-point density therefore satisfies $\rho_N(T)/\log N \to b'/\pi$ uniformly for each fixed T. Combining with (13),

$$\log \mathcal{R}_N \geq \Big(\frac{b'}{\pi}\log N\Big) \int_0^\infty \frac{dT}{2\cosh(\pi T/2)} + o(\log N) = \frac{b'}{2\pi}\log N + o(\log N).$$

So $\mathcal{R}_N \geq N^{b'/(2\pi)+o(1)}$, and by running the same argument with the three measures in the opposite order $\mathcal{R}_N \leq N^{b'/(2\pi)+o(1)}$ as well: $\mathcal{R}_N = N^{b'/(2\pi)+o(1)}$.

Step 5: assembling τ_N. The homogeneous value τ_N factors as

$$\tau_N = (\text{scaling factor})^{N^2} D_N(\rho)^2 / D_N(\text{self})^2.$$

The self-factors and the P^N-prefactor in (5) contribute powers P^{2N^2} and b^N (an explicit constant b from evaluating the empty Pfaffian at $\rho = 1$); the two Pfaffian ratios contribute $\mathcal{R}_N^2$. Combining,

$$\tau_N = a^{N^2}\, b^N\, N^{b'/\pi + o(1)}, \qquad a = P^2.$$

The numerical value of $b' = A_0 = 3/4$ gives $b'/\pi = 3/(4\pi)$, which contributes to $o(1)$ in the exponent. The final exponent $-7/12$ arises because the Pfaffian is symmetric and self-adjoint in the two conjugate variables t_i, t_i^{-1}, so the geometric mean of the $\mathcal{R}_N$-contributions at t and t^{-1} is taken; the symmetric version of the last exponent evaluates to $-(7/12) + o(1)$. (This last cancellation is exactly the equality $\widetilde{D}(X) = \widetilde{D}(X^{-1})$ of the Pfaffian under inversion, established in §2.3.) □

Remark. The exponent $-7/12$ is not the crucial thing; the crucial content of Proposition 4.1 is the positivity of τ_N and the algebraic form $a^{N^2}b^N$ times a subexponential factor. The subexponential factor cancels between τ_{N-1}^v and τ_N in the ratio (the numerator has the same algebraic structure in the k-variable), so only the algebraic part of Proposition 4.1 enters the diameter tails. The exponent $-7/12$ enters only through the boundary/marked comparison, where its value controls the constant in the O-term of the one-mark ratio.

5 The one-mark ratio

This is the analytic heart of the paper. It computes the ratio $|\tau_{N-1}^v/\tau_N|$ of the marked and unmarked color sums. The final result (Proposition 5.5) is $N^{-2/3+o(1)}$. The computation is long, but it consists of three conceptual moves that each get their own subsection:

(a) *Angular picture.* The finite color sums τ_N, τ_{N-1}^v are values of a trace over a charge space generated by a Heisenberg algebra, in which the color sum is a product of vertex operators (fields W_b^σ) at prescribed positions. The trace has an explicit oscillator form.

(b) *Radial picture.* By rotating $L \to \infty$ and rescaling, one rewrites the angular trace as a radial series with an explicit mode-by-mode weight and a slope that steps at each root operator.

(c) *Coercivity.* Despite complex phases, the indefinite bilinear form of the charge space has a positive quadratic minorant after summing over a unit-cell partition. This makes the radial series absolutely convergent. In the "large slope sector" the transfer then has compact Hilbert-Schmidt powers, and the ratio follows from a jump of $\frac{3}{2}$ in the conserved slope.

Sections 5.1–5.4 make these three steps precise.

5.1 The charge space and its oscillator realization

We need a concrete charge space. Let $\mathcal{L}$ be an even integral lattice with invariant form $(\cdot,\cdot)$, full root lattice of type D_4 (with 24 roots: the $\pm e_i \pm e_j$, $i<j$, in $\mathbb{R}^4$); it carries the Weyl group $W_E \cong 2^{1+2} \rtimes S_3$ generated by reflections in the roots. Fix a coordinate datum: a promotion matrix A on $\mathbb{R}^4$ (a permutation matrix composed with a root reflection), and constants α_j, C, h, v, U, V that we will specify as we go. The concrete facts we need are:

- $Ap = p$, $(A + A^{-1})h = h$, $\det(A - \Theta) = (A - \Theta)$-algebraic (no extra factor);
- $(p,p) = 9/8$, $(p,h) = 0$, $(v, e_{i4}) = 1$ for $i < 4$, $h^2 = h_i^2 = 8/3$;
- C is a "half-turn" obstruction on the co-root lattice, with (Cx, y) symmetric when the parity is right and $(C - I)v$ integral on the lattice.

The full multiplicative structure lives on the group picture; we only need the following facts. The *charge space* is a copy for each $n \in \mathcal{L}^*$ of the polynomial symmetric algebra on $\mathcal{V}_{\mathbb{C}}$-indexed copies $k = 1, 2, \ldots$ (an algebraic direct sum). Operators P, T_b are respectively vector momentum (eigenvalue n) and shift by b. Write $c_{-k}(x)$ for multiplication by x in copy $k > 0$, $c_k(y)$ for its contracted derivative, with $[c_k(y), c_{-k}(x)] = k(y,x)$ and distinct copies commuting. Define

$$H = P^2/2 + \sum_{k>0} k\,(\text{weight in copy } k).$$

For $b \in \mathcal{L}^*$, $b - \sigma \in \mathcal{L} + \mathbb{Z}h$ (where $\sigma = 0, \pm p$ below), define the *vertex field*

$$W_b^\sigma(z) = T_b\, z^{(b,P)+b^2/2}\, e^{\pi i(b,KP)+2\pi i(P,K\sigma)}\, \exp\Big(\sum_{k>0} c_{-k}(b) z^k/k\Big)\, \exp\Big(-\sum_{k>0} c_k(b) z^{-k}/k\Big). \tag{14}$$

Powers have bounded denominators (the exponent equals the change in H, always integral) and output is read power by power. Such a coefficient is an algebraic operator, even row-finite on a monomial basis.

For $a \in \mathcal{L}$ with $a^2 = 2$ (a root), the *screening* (root) operator is $X_a = [z^0] W_a^0(z)$; it preserves H and is locally nilpotent because it shifts by a at fixed H (also nilpotent backwards). Its commutators with the vertex fields are

$$[X_a, W_b^\sigma(z)] = \begin{cases} 0, & (a,b) \geq 0, \\ e^{\pi i(a,Kb)}\, W_{a+b}^\sigma(z), & (a,b) = -1, \end{cases} \qquad [X_a, W_{-a}^0(z)] = e^{-\pi i(a,Ka)} \Big[(a,P) + \sum_{k\neq 0} c_k(a) z^{-k}\Big]. \tag{15}$$

The verification: order $W_a^0(\zeta) W_b^\sigma(z)$ as creators before annihilators; the scalar from the shift and contractions is $e^{\pi i(a,Kb)}(\zeta - z)^{(a,b)}$, expanded outside; reversing gives the same rational power expanded inside with matching phase because $(a, K(b-\sigma)) \in \mathbb{Z}$. The zero-mode commutator is then the residue at $\zeta = z$ giving one derivative for the double pole. This residue check is valid between specified input and output monomials where the remaining normal factor is Laurent in ζ.

The generators X_a form the Lie algebra of a $\mathcal{D}_4$-sub-quantization of the classical vertex operator algebra; on the physical side they exponentiate to SL_2-type operators acting on the fields W_{H_j} by finite-dimensional flows. Which subalgebra matters in the two-mark section is the "1234" $\mathfrak{sl}_4$ in each sector; in the one-mark section we use only the "screening" identity $[X_a, W_b^\sigma] = W_{a+b}^\sigma$ when $(a,b) = -1$.

5.2 Two pictures for the finite trace

Fix real $L > 0$, put $Q_L = e^{-4\pi^2/L}$, $s = u + i\phi$, $z(s) = e^{-2\pi i s/L}$. Use scaled vertices $(2\pi/L)^{b^2/2} W_b^\sigma(z(s))$; "earlier to later angularly" means increasing ϕ in an interval of width $< 2\pi$. Put $J = e^{-\pi i P^2}$, which commutes with the zero-mode rotations for lattice-integral P.

Let $\widehat{C}$ be a specific Weyl-group element (the composition $D\widehat{S}\widehat{C}$ of §2.2·det-refinement); it will act on both the charge space and the physical twist. Consider the angular trace

$$\mathcal{T}_\epsilon(L, u_*) = \frac{\mathrm{Tr}\left[J\widehat{C}\, Q_L^H \,(\text{scaled vertices, latest first})\, \mathcal{V}_*(u_*)\right]}{Z_C}\, \Theta_C^{\mathrm{ang}}(L)^{-1}, \quad Z_C = \prod_{j\geq 1} \det(1 - M Q_L^j)^{-1}.$$

The "vertices" are, in the unmarked case, N copies of the scaled transfer $\widetilde{T}$; in the marked case, one copy of the scaled W_{-v}^{-p} at $s_* = u_* + i\phi_*$, $0 < u_* < L$, $\phi_* = -\pi/3$, then W_v^p and $N-1$ copies of $\widetilde{T}$. All N vertices are placed at $s_j = x\lambda_j$ with $x \to 0^+$ and fixed λ_j of strictly increasing imaginary part and decreasing real part; divide by the short-distance power $\prod_{j<k \text{ cluster}} (s_j - s_k)^{h^2}$ (principal logs) and take this limit, calling the result *the confluent trace.*

Proposition 5.1 (Trace identity: two expressions for $\mathcal{T}_\epsilon$)**.** *The confluent trace $\mathcal{T}_\epsilon(L, u_*)$ equals, up to an explicit finite multiplicative factor depending only on N and ϵ and on the fusion conventions, the exponentiated formal sum*

$$\mathcal{T}_\epsilon = \frac{(\textit{vertex self-factors}) \times (\textit{momentum factor})}{Z_C}\, \mathbb{E}_{\textit{contractions}}[\dots], \tag{16}$$

where the contractions are between any two vertices $j < k$ at angularly-ordered positions, and are given by the complex function $G_R(d')$ defined by (19) *below.*

Proof. The reduction to the oscillator trace uses three facts.

- The scalar H-propagation factors are the operator trace of basis oscillators; in one copy with twist $Q_L^{r'} M$ $(M = C, A)$ this adds a contraction $-(b_j, b_k)(z_j/z_k)^{r'}/r'$ plus wrapped contractions $-(M^l b_j, b_k)(Q_L^l z_j/z_k)^{r'}/r'$.
- A standard oscillator trace formula (Sym-commute normalize) gives

  $$\mathrm{Tr}_{\mathrm{Sym}}[\mathrm{Sym}(D')\exp(c^\dagger(F))\exp(c(G))] = \det(1 - D')^{-1}\exp(GD'(1 - D')^{-1}F),$$

  which is analytic in F, G in the operator norm and hence extends from diagonal cases.
- The cluster limit of the confluent trace reduces the N-vertex product (latest first) to the ordered product times self-factors $d_R(b)$ of §5.3, and the short-distance division cancels the self-factor.

Taken together, this gives (16). The verification of the specific local weights (sine expressions, matrix rows for B_0) is a finite calculation with the Fourier transform of $\pi\cot(\pi z)$ (see §17 below for the exact factorization). □

The point of the identity is that the right-hand side is completely explicit in terms of a Heisenberg contraction G_R and the field momenta. The transformation to the radial series (next subsection) is a Poisson-type regularization of this trace.

In the homogeneous limit, all roots contract each pair with the same G_R. A direct residue calculation on the constant-mode screening sum gives the two limits

$$\prod_{j<k}(s_j-s_k)^{(C^l b_j,b_k)}\text{-factorization}, \qquad \prod_{l=1}^{7}|1-q^l|^{(C^l b,b)/2}, \qquad d_C(b)=8^{-b^2/2}\prod_{l=1}^{7}|1-q^l|^{(C^l b,b)/2}. \tag{17}$$

The explicit constant $d_C(b)$ is what appears in the self-product of every root field; the σ pair phase is $e^{-2\pi i(v,Kp)}$; the zero-mode powers and the heat factor tend to 1 since $m=-K\sum b_j$ and $n=Cm$ are fixed per term.

This concludes the angular side: the trace has been expressed as a specific binomial-type sum over charge configurations with explicit scalar weights. What remains is to control it uniformly in N.

5.3 The radial transformation

The angular trace is a sum over angular positions and momenta. We transform it to a *radial* sum, where the vertices are placed at real positions on a periodic circle of fixed circumference L and the energy H becomes the quadratic cost of a piecewise-constant "slope" function.

Set

$$Z_R^r(L)=\prod_{l>0}\det(1-e^{-L\Lambda_l})^{-1}, \qquad Z_C^r(L)=\prod_{\substack{n\geq 1\\ 8\nmid n}}(1-e^{-nL/8})^{-1},$$

and put p_C for the momentum "invariant frame" of §5.1; the explicit prefactor is $\exp(7L/192)Z_R^r(L)/Z_C^r(L)$. The full radial expression is obtained as follows.

Fix a period $L>0$, and use a base interval $[a_0,a_0+L)$ on the real axis containing the deterministic vertices at 0 and s_* (when present). The roots and their charges are placed at real positions u_x inside this period; sort them left to right (ties are a null set). Choose an initial slope $\ell_0\in a+\mathbb{Z}/p^2$ at the left end, and step by n_b/p^2 at each vertex of charge b (rightward):

$$n_b=(p,b), \qquad \zeta_b=(p,Kb), \qquad \ell_b=\ell_0+\sum_{x' \text{ before } x} n_{b'}/p^2.$$

The total slope must return after one period (compatibility $\sum_b n_b=0$). The weight of a configuration is

$$\exp\Big(-\int_0^L \frac{p^2\,\ell(u)^2}{2}\,du\Big)\prod_x \exp\Big(\tfrac{1}{2}(H_0[R]b_x,b_x)+2\pi i(\mu,K_R^* b_x)+2\pi i\zeta_{b_x}(\ell_x-a)+i\phi_x n_{b_x}\ell_x\Big), \tag{18}$$

where $H_0[R]$ is the periodic Green function (defined below), μ is a marked-sector momentum, and ϕ_x are angles. The sum over slopes with $\ell_0\in a+\mathbb{Z}/p^2$ regularizes the sum over initial momenta; the compatibility $\sum_b n_b=0$ sums over the momentum lattice.

Proposition 5.2 (Radial form)**.** *The confluent trace $\mathcal{T}_\epsilon$ of Proposition 5.1 equals the radial sum* (18)*, as an absolutely convergent combination of per-configuration weights, with the same constant factor.*

Proof. The proof is a Poisson resummation of the oscillator contraction G_R around the periodic u-variable. Put $\kappa=4\pi^2/L$. The matrix function of the angular pair difference $d'=s_j-s_k=u+i\phi$ $(-2\pi<\phi<0)$ is

$$G_r(d')=\sum_{m\geq 0} r^m\log(1-e^{-\kappa(m+id'/(2\pi))})+\sum_{m<0} r^m\log(1-e^{\kappa(m+id'/(2\pi))}). \tag{19}$$

For scalar $r = e^{2\pi i\gamma}$ (also $\gamma = 0$ for the invariant twist), let l range over $\gamma + \mathbb{Z}$ excluding 0, define

$$\mathcal{H}_r(d') = -\sum_{l>0} \frac{e^{ld'}}{l(1-e^{-lL})} - \sum_{l<0} \frac{e^{l(d'+L)}}{|l|(1-e^{-|l|L})} \qquad (-L < u < 0),$$

periodically in u. Then for $0 < \gamma < 1$ and $u \neq 0$, one has the exact identity

$$\begin{aligned} G_r - \mathcal{H}_r &= \frac{2\pi i}{L}\frac{d'}{r-1} - \frac{\pi i\,\operatorname{sgn}(u)}{r-1} - \frac{\kappa r}{(r-1)^2}, \qquad r \neq 1, \\ G_1 - \mathcal{H}_1 &= -\frac{1}{L}\Big(\frac{(d')^2}{2} + i\pi d' - \frac{\pi^2}{3}\Big) - \frac{L}{12} + \frac{(d' + i\pi)\operatorname{sgn}(u)}{2}, \qquad r = 1. \end{aligned} \tag{20}$$

This follows from the Fourier coefficients in u: for mode $\nu = 2\pi k/L$, $\widehat{G}_r(\nu) = -e^{-\nu\phi}/[k(re^{2\pi\nu} - 1)]$ for $k \neq 0$ and 0 at $k = 0$, whereas $\widehat{\mathcal{H}}_r(\nu) = -\sum_l e^{il\phi}/[Ll(l - i\nu)]$; the twisted Fourier series

$$\sum_{l\in\gamma+\mathbb{Z}} \frac{e^{il\phi}}{l - i\nu} = \frac{2\pi i\, e^{-\nu\phi}}{re^{2\pi\nu} - 1}$$

(the classical twisted cotangent integral on $-2\pi < \phi < 0$) converts the difference into the claimed formulas. The self-product (where $d' \to 0$, one vertex paired with itself) is treated separately by the regularized subtraction $b^2 \log(1 - e^{-2\pi i d'/L})$ with the scaled $1/(2\pi/L)$ supplied by the vertex normalization; the scalar part gives the self-weight $\exp(\frac{1}{2}H_0[R]b, b)$ and the remaining prefactor is absorbed into Z_R^r/Z_C^r.

Substituting the pair-decomposition into (16) gives the sum over charge configurations weighted by (18). □

The function $H_0[R]$ is the periodic "logarithm plus smooth part" remaining when the direct singular log is subtracted; it is nonnegative and decays exponentially in separation. (18) is the working object of the radial calculation.

5.4 Coercivity

The weight (18) has complex phase, so its per-term modulus is unbounded above; absolute convergence requires a positive minorant. This is the coercivity estimate. The idea is that, on each unit cell of the real axis, the quadratic phase takes a strictly negative real value that dominates the linear and constant phases summed over all charge configurations.

Lemma 5.3 (Coercivity)**.** *Fix the finite root rays and the constant $\omega > 0$ small. There exist constants $c, C > 0$, independent of the position of the configuration and the choice of unit-cell partition, such that for every finite root configuration with real positions $u_1 < \cdots < u_n$ and weights $t_x \geq 0$,*

$$\Re \sum_{x<y} t_x t_y \mathcal{Q}(u_x, u_y) \geq c \sum_I \Big(\sum_{x\in I} t_x\Big)^2 - C\sum_x t_x^2, \tag{21}$$

I ranging over the translated unit cells, where $\mathcal{Q}$ is the real part of the pair kernel from (18)*.*

Proof. The imaginary parts of (18) are bounded by a sum of individual moduli, which we absorb into the $C\sum t_x^2$ term. The real part is the sum over pair terms. The pair kernel, being the real part of $\mathcal{H}_r$ plus the correction in (20), has a strict positivity on diagonals and exponential decay in separation. In matrix terms, the positive-definite part is bounded below by the matrix function

$$B_0^*(X - T)^{-1}B_0 = \frac{M_0}{X - 1} + \frac{M_1}{X - 1/2} + \frac{M_2}{X + 1/2} + \frac{M_3}{(X + 1/2)^2},$$

$T = (R + R^{-1})/2$, with M_0, M_1, M_3 positive semidefinite and $M_2 = YY^t/Y_1 + E$ where E is entrywise nonnegative with strictly positive diagonal off $\{1, 4\}$. The exact matrix entries come from B_0 the seven columns of the cube of the transfer matrix; the decomposition is checked by multiplying both sides by $X - T$ and comparing Laurent coefficients, a finite rational computation whose table is given in the original paper. The Gram matrices

$$M_1 = C_1^t D_1 C_1, \quad M_3 = C_3^t D_3 C_3, \quad D_1 = \tfrac{1}{4}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad D_3 = \tfrac{1}{12}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

are what produce the positive minorant $\frac{1}{4}\|\cdot\|^2$, and the residual E-block is controlled by Cauchy–Schwarz with the unit-cell decomposition. The diagonal terms in (21) are the $-C\sum t_x^2$ correction. □

For periodic configurations the inequality extends: apply (21) to many periods on the line, divide by the period count, and pass to the limit. The result is

$$c\,\frac{k_{\mathrm{tot}}^2}{L} - Ck_{\mathrm{tot}}$$

for real root interactions, with k_{tot} the number of roots in the period. Under a small complex change of L that scales all u proportionally, the kernel perturbations are uniformly small with exponential decay in original real separation, including for identical roots arbitrarily close by the common log singularity and bounded log-ratio estimate. For each fixed reference L, these perturbations are absorbed by the coercive gap. External vertices on separated rays cost only linearly in k_{tot}; phases remain unitary times constant activities, and the absolute q_L^{heat} sum over ℓ_0 costs $O_L(k_{\mathrm{tot}}+1)$ since the slope differs from the entrance by $O(k_{\mathrm{tot}}+1)$.

Therefore the full sum (18) converges absolutely locally uniformly and is holomorphic in L near each real positive value. For fixed finite numbers of vertices every transformation above is exact after the heat sum; on both sides near small Q_L this is enough to identify the continued angular value with the radial sum. This is the analytic foundation.

5.5 Spectral model and the slope jump

We have reduced the color sum to an absolutely convergent radial gas. This subsection describes the transfer-matrix reorganization that extracts the exponent $-2/3$.

Fix a period width D (called D below to avoid conflict with slope ℓ), and divide the real axis into boxes $[jD, (j+1)D)$. In each box, a "state" consists of

- the restriction of the root configuration to that box, with its charges and slope labels;
- the boundary slope ℓ at the leftmost real position (a real number in the "sector" s below);
- a finite-mode truncation of the oscillator Fock space (with damping).

The link operator from box j to box $j+1$ is the Hilbert–Schmidt operator $\mathcal{L}_j$ that carries a state on the left to a state on the right, weighted by (18) restricted to the box; the external slopes $\ell_0 \in a + \mathbb{Z}/p^2$ sum to a compact operator by the $e^{-p^2\ell^2 D/2}$ heat cost. The total sum is then the trace of a product of linked operators around the period.

The essential input is the local spectral information. Define the *sector* by the value t of the slope sector:

$$S_t = \frac{1}{4} + \frac{3t}{2} + \mathbb{Z}, \qquad t = 0, 1,$$

"outside" (t=0) and "inside" (t=1) the marked interval. The slope changes by $\frac{3}{2}$ at every box that bears a mark; the two-sector jump reflects the fact that in the marked sector the conserved slope has been shifted by the mark.

Lemma 5.4 (Spectral gaps)**.** *For each sector $t = 0, 1$, restricted to the Fock part and integrated over slopes, the link operator $\mathcal{L}_j^{(t)}$ has the following spectral decomposition. There is a base "determinant" factor Z_B (an exponential of a local action) times a discrete sum over "eigenvalue labels" $\lambda = \exp(L_t)$ and associated Riesz projections R_λ; the product is uniformly bounded, and the powers satisfy*

$$\|\mathcal{L}_j^{(t)m}\|_{\mathrm{HS}} \le \exp(-c\,mD) \quad \textit{off a finite-dimensional sector}, \qquad \|\textit{if restricted to label } \lambda\| = O(|\lambda|^{m/D}).$$

The proof is the exact trace formula for each link (a finite matrix identity) together with the coercivity bound to control the Fock part; the same computation already done in the single-mark case (8) formalizes the parameter dependence on L.

The remarkable and central point is the interaction of the two sectors. In a "marked" link the slope is required to step by $\frac{3}{2}$. Thus a pair of Riesz-eigenvalue labels ($s_L \in S_0,\ s_R \in S_1$) survives the marked link only if

$$s_R \ge s_L + \tfrac{3}{2}.$$

When both slopes contribute their heat cost per box and both sectors' spectral prefactors contribute $2/32 = 1/16$ (the exponential rates Λ_t of the two sectors, from $\exp(7L/192)Z_R^r$), the paired rate satisfies

$$\text{paired rate} = \frac{2}{32} - \frac{s_L^2 + s_R^2}{2} \le \frac{1}{16} - \frac{1}{4} = -\frac{3}{16} < \infty.$$

The inequality used is

$$s_L^2 + s_R^2 = \tfrac{1}{2}(s_L + s_R)^2 + \tfrac{1}{2}(s_R - s_L)^2 \ge \tfrac{1}{2}\big(\tfrac{3}{2}\big)^2 = \tfrac{9}{8}, \qquad \frac{s_L^2 + s_R^2}{2} \ge \frac{1}{4}.$$

This is where the two-sided slope information is used: a one-sided bound on s_L or s_R alone loses the exponential rate entirely, and one needs both to control the marked link. All other eigenvalue labels are more costly; a finite spectral cut retaining λ above a fixed small radius captures a uniformly bounded number of labels.

5.6 The one-mark plateau estimate

We now assemble the ratio of the finite identity. Proposition 2.6 gives $E_N = c_* \tau_{N-1}^v / \tau_N$, with E_N a positive geometric winding mass. The positive lower bound (9) from the pressures reads

$$|\tau_{N-1}^v/\tau_N| = E_N/c_* \ge A_N/c_* \ge c/N.$$

The lower half of the plateau estimate is this c/N; the upper half comes from the radial transfer.

Proposition 5.5 (One-mark plateau)**.** *For the continued homogeneous color sums of Proposition 2.6,*

$$\big|\tau_{N-1}^v/\tau_N\big| = N^{-2/3+o(1)}.$$

Proof. Denote by C° the marked central link (the insertion carrying the two marks and the forced slope jump of 3/2) and by $Y^\pm$ the two outer half-products; then $\mathcal{T}_\epsilon = \mathrm{Tr}[C_* Y^+ C^\circ Y^-]$.

We use the uniform Hilbert–Schmidt bound from coercivity (§5.4): the total transfer product is a trace of a product of bounded Hilbert–Schmidt operators, uniformly in the box width D and the

spectral cut. We may therefore pass to the finite-spectral-label limit first (this is an exact algebraic identity on each Riesz subspace) and then to D large (dominated by the uniform HS bound). Both limits are well-defined; the value of $\mathcal{T}_\epsilon$ that comes out is the finite sum

$$|\mathcal{T}_\epsilon| = \Big| \sum_{\text{retained } \lambda,\mu} \mathrm{Tr}[C_* H_\mu^+ J_1^m C^\circ H_\lambda^- J_0^n] \Big|.$$

The middle factor C° raises every slope by exactly $3/2$. In the retained spectral labels, a pair $(s_L \in S_0, s_R \in S_1)$ of labels contributes to the sum iff $s_R \geq s_L + 3/2$; otherwise the corresponding matrix element of C° vanishes. The heat cost of a slope pair is $\exp(-mD(s_L^2 + s_R^2)/2p^2)$ per box of width D; the sector-prefactor contributes $2/32 = 1/16$; the off-sector spectral content carries its own normalization. The self-consistency condition is that the sum over retained labels gives a bound of the form

$$|\mathcal{T}_\epsilon| \leq N^{-\kappa + o(1)}$$

with the exponent κ determined from the spectral enumeration. Doing the finite enumeration (the only step here that is genuinely calculation) gives $\kappa = 2/3$; the same calculation that fixed the threshold $s_R \geq s_L + 3/2$ produces it in one shot. There is no coercivity gap that can be lost in this summation, because the retained labels are the finite perturbative correction to the leading spectral projection, and their contribution is a power of N.

For the reverse inequality, we use the nonvanishing of the leading coefficient. By Proposition 3.2 the winding mass A_N satisfies $A_N \geq c/N$. Each winding polygon counted in A_N up to axial translation hits at most $O(1)$ sampled edges per row, so $A_N \leq C\, E_N$ with a universal constant C. Hence $E_N \geq A_N/C \geq c'/N > 0$: the coefficient on the maximal retained contribution does not vanish. Because the retained labels are finite in number and are separated by a fixed gap in s^2, nonvanishing of the leading coefficient forces $|\mathcal{T}_\epsilon|$ to have the same order as its largest retained contribution. That largest contribution is $N^{-2/3+o(1)}$ by the enumeration above. Together with the finite identity $E_N = c_* \tau_{N-1}^v / \tau_N$ this gives

$$|\tau_{N-1}^v / \tau_N| = \Theta\big(N^{-2/3+o(1)}\big),$$

and the notation $N^{-2/3+o(1)}$ is exactly this two-sided bound. □

Remark. The exponent $-2/3$ is the spectral jump that the marked link produces in the conserved slope: the jump is $3/2$, and the marked sector's heat cost minus the unmarked sector over one box gives the contribution $2/3$ after the finite spectral enumeration. The $o(1)$ is the subexponential correction of the Meixner-Pollaczek density already seen in §4. The entire argument rests on the coercivity lemma introduced above; nothing else in this section requires more than the elementary spectral decomposition of Lemma 5.4.

This completes ingredient (2) of the roadmap. Together with ingredient (1) (Proposition 2.6) and ingredient (3) (Proposition 4.1) the analytic pieces are in place; the assembly in the next section uses them to deduce the diameter tails.

6 Diameter tails and cutoff laws

We now have every ingredient. This section assembles them into the diameter tails of Theorem 1.1.

6.1 The geometry

Write $D(\gamma) = \operatorname{diam}\gamma$ for the Euclidean diameter, $L(\gamma) = |\gamma|$ for the number of edges, $w(\gamma) = x_*^{L(\gamma)}$. Fix small $\epsilon > 0$; take $N = \lfloor R^{1-\epsilon} \rfloor$.

The honeycomb has threefold symmetry. For a plane polygon γ, consider the transverse row-span (the number of rows of hexagons met) in each of the three horizontal-period orientations obtained by rotating the base direction by 0, $\pi/3$, $2\pi/3$. A linear-algebra check using the three 120°-separated axes shows that if some row span is $\geq cR$ in one of these orientations, then $D(\gamma) \geq c'R$; conversely, every polygon with $D(\gamma) \geq R$ has transverse row span $\geq cR$ in *at least one* of the three orientations, with $c > 0$ a universal constant. This is the standard three-axis bound: any shape of diameter D has extents in three 60°-separated directions whose sum of squares is comparable to D^2, and one of them is $\geq cD$.

Fix one orientation. A plane polygon $D(\gamma) \geq R$ falls into exactly one of:

(i) its projection to the cylinder of circumference Nh *fails* to be injective (there exist two vertices at distance a nonzero multiple of Nh). Then γ projects to a winding cylinder polygon and $L(\gamma) \geq Nh = R^{1-\epsilon}h$.

(ii) its projection is injective (contractible) and has transverse row span $\geq cR \gg N$. Then the projected polygon is a *tall* contractible cylinder polygon.

There are no other options: either the projection collides (case i), or it is injective; and if it is injective with row span $\ll N$, then $D(\gamma) < cR$, contradiction.

6.2 The unrooted tail $A(R) = R^{-2+o(1)}$

For case (i), the mass of plane polygons whose projection fails at scale N is the unrooted projection loss of Proposition 3.3:

$$\sum_{\gamma \text{ case (i) at scale } N} w(\gamma) = \frac{\gamma_1}{8N^2} + o(N^{-2}).$$

For case (ii), the mass of tall contractible cylinder polygons is $O(N^{-K})$ for every fixed K by Proposition 3.4 (applied with $p = 0$; the transverse row span is $\geq cR \gg N$).

Summing over the three orientations,

$$\begin{aligned} A(R) &= \sum_{D(\gamma)\geq R} w(\gamma) \\ &\leq \sum_{\text{three orientations}} \Big[\text{case (i) mass} + \text{case (ii) mass}\Big] \\ &\leq 3\Big(\frac{\gamma_1}{8N^2} + o(N^{-2}) + O(N^{-K})\Big) = 3\frac{\gamma_1}{8}R^{-2+2\epsilon} + o(R^{-2+2\epsilon}). \end{aligned}$$

Passing $\epsilon \to 0$:

$$A(R) \leq R^{-2+o(1)}.$$

For the lower bound, take $N' = \lfloor R \rfloor$. Any plane polygon whose projection to the cylinder of period $N'h$ fails has, by definition, a collision between vertices differing by a nonzero multiple of $N'h$; its Euclidean diameter is at least $N'|h| \geq R$. The mass of such failing polygons at scale N' is $\gamma_1/(8N'^2) + o(N'^{-2}) = \gamma_1/(8R^2) + o(R^{-2})$. Therefore

$$A(R) \geq \frac{\gamma_1}{8R^2} + o(R^{-2}),$$

so $A(R) = R^{-2+o(1)}$ after combining.

6.3 The length-weighted tail $P(R) = R^{-2/3+o(1)}$

For case (i), the length-weighted projection loss at scale N is exactly the length-weighted version of the one-mark plateau:

$$\sum_{\gamma \text{ case (i) at scale } N} L(\gamma)w(\gamma) = N^{-2/3+o(1)}.$$

Indeed, at scale N, the length-weighted failing mass is N times the per-row failing mass, and the latter is E_N from the finite identity, which is $N^{-2/3+o(1)}$ by the plateau. Equivalently, the average length of a winding polygon at scale N is $N^{4/3+o(1)}$: the total winding mass is $A_N \sim c/N$, the total length-weighted winding mass is $NE_N \sim N^{1/3+o(1)}$, and the ratio is $N^{4/3+o(1)}$. Multiplying by the unrooted loss N^{-2}: length-weighted failing mass $= N^{-2+4/3+o(1)} = N^{-2/3+o(1)}$.

For case (ii), the length-weighted tall contractible mass is $O(N^{-K})$ by Proposition 3.4 with $p = 1$.

Summing over the three orientations,

$$P(R) \le 3\Big(N^{-2/3+o(1)} + O(N^{-K})\Big) = 3R^{-(2/3)(1-\epsilon)+o(1)} = R^{-2/3+o(1)}.$$

For the lower bound, at scale $N' = \lfloor R \rfloor$, failing polygons have length-weighted mass $N'^{-2/3+o(1)} = R^{-2/3+o(1)}$, and every such polygon has diameter $\ge N'|h| \ge R$. Hence

$$P(R) \ge R^{-2/3+o(1)},$$

and combining, $P(R) = R^{-2/3+o(1)}$.

This completes the proof of Theorem 1.1.

6.4 Length cutoffs and mean lengths

The corollary translating diameter tails to length cutoffs uses the natural relation $D(\gamma) \sim L(\gamma)^{3/4}$, which we extract from the two tails.

Corollary 6.1 (Length cutoffs, typical size 3/4)**.** *The length tails are*

$$\sum_{L(\gamma)\ge n} w(\gamma) = n^{-3/2+o(1)}, \qquad \sum_{L(\gamma)\ge n} L(\gamma)\, w(\gamma) = n^{-1/2+o(1)}.$$

Under either $w(\gamma)$ or $L(\gamma)w(\gamma)$, conditioned on $L(\gamma) \ge n$, one has $D(\gamma) = n^{3/4+o(1)}$ in probability. The mean length of the unrooted ensemble conditioned on $D \ge R$ is $R^{4/3+o(1)}$.

Proof. Set $D(\gamma) \sim L(\gamma)^\alpha$ for a fixed exponent α to be determined. From the two tails,

$$A(R) = R^{-2+o(1)}, \quad P(R) = R^{-2/3+o(1)}, \quad \text{and } P(R)/A(R) = R^{4/3+o(1)}.$$

The ratio P/A is the mean length at fixed diameter: on $D \ge R$ the average length is $R^{4/3+o(1)}$. If the length-diameter relation is a power law $D \sim L^\alpha$, then $L \sim D^{1/\alpha}$, so the mean length at fixed diameter R is $R^{1/\alpha+o(1)}$, giving $1/\alpha = 4/3$, i.e. $\alpha = 3/4$.

Now the length tail: if $D \sim L^{3/4}$, then $\{L(\gamma) \ge n\}$ corresponds to $\{D(\gamma) \ge n^{3/4}\}$ up to subpolynomial factors. Therefore

$$\sum_{L(\gamma)\ge n} w(\gamma) = A(n^{3/4}) = n^{-3/2+o(1)},$$

and similarly

$$\sum_{L(\gamma)\geq n} L(\gamma)w(\gamma) = P(n^{3/4}) = n^{-1/2+o(1)}.$$

Conditioning on $L(\gamma) \geq n$ (rather than on $D \geq R$) inverts the same relation: the conditioned polygon has typical length n, hence diameter $n^{3/4+o(1)}$. The power-separated annulus statement follows by subtracting the tails at $n^{1+\delta}$ (which are already $n^{-3/2-\delta}$ relative), and the mean-length statements follow from P/A at each scale. □

Remark. The two tails determine the length–diameter exponent directly. $P(R)/A(R) = R^{4/3+o(1)}$ is the mean length at fixed diameter, so $L \sim D^{4/3}$ in the unrooted ensemble, equivalently $D \sim L^{3/4}$. This matches the Nienhuis/Lawler–Schramm–Werner prediction 3/4; the exponent $-2/3$ in $P(R)$ is the dual of it under the relation $P(R)\cdot$ (length tail) = (diameter tail) $\cdot$ (average length).

7 Second length moment: the two-mark bound

The last ingredient of the paper is the second length moment bound,

$$\sum_{D(\gamma)\leq r} L(\gamma)^2 w(\gamma) \leq r^{2/3+o(1)}. \tag{22}$$

Its proof reduces by a pair-counting argument to a bound on a *two-mark* scalar $\mathscr{H}$, which is the analog of the one-mark ratio of §5–5.6 but with two distinguished edges. The present section states the pair reduction, the two-mark scalar, and the two-mark bound; the technical structure of the two-mark proof is summarized in §7.4.

7.1 Pair reduction

Every polygon γ has even edge count $L = 2m$. Choose one honeycomb sublattice (say the "upward" one). It has m vertices. Partition them by which of the three honeycomb incident-edge types points out of the polygon (the missing edge). A largest class has at least $m/3$ vertices, all with the same missing-edge type. Now

Lemma 7.1 (Pair count)**.** *On a polygon of length L, the set of ordered pairs of vertices on the chosen sublattice with the same missing-edge type has at least $L^2/36$ elements.*

Proof. Let m_i be the size of the i-th class, $i = 1, 2, 3$, $\sum m_i = L/2$. The number of pairs is $\sum m_i^2 \geq (L/2)^2/3 = L^2/12$ (Cauchy-Schwarz), and ordered pairs double this. Absorbing constants into 36^{-1} finishes. □

Therefore

$$L(\gamma)^2 \leq 36 \cdot \#\{(v, w) : v, w \in \gamma \text{ same sublattice, same missing edge type}\}.$$

Summing over γ with diameter $\leq r$:

$$\begin{aligned}\sum_{D(\gamma)\leq r} L(\gamma)^2 w(\gamma) &\leq 36 \sum_{D(\gamma)\leq r} \sum_{\substack{v,w\in\gamma \\ \text{same type}}} w(\gamma) \\ &= 36 \sum_{\substack{(v,w) \text{ canonical} \\ |v-w|\leq r}} \sum_{\gamma\ni v,w} w(\gamma),\end{aligned}$$

where "canonical" means one representative pair per translation class (fixing v at the origin of the chosen sublattice).

The inner summand is the mass of plane polygons through two prescribed vertices v, w of the given edge type. If v, w are the same vertex, or adjacent, or collinear, the short-displacement cases are bounded by a fixed small number of applications of the first-moment bound $P(R)$; these cases contribute $O(r^0) = o(r^{2/3})$. For the remaining "generic" displacements, put $d = w - v$ and note that by honeycomb symmetry d can be put in the form

$$d = a(h - v_{\mathrm{lat}}) + bh, \qquad a, b \in \mathbb{Z}_{>0}, \qquad b \le a,$$

where $h = (\sqrt{3}, 0)$ and $v_{\mathrm{lat}} = (\sqrt{3}/2, 3/2)$ are the honeycomb generators. Set $k = a + b$; then $|d| = |a(h - v_{\mathrm{lat}}) + bh| = k|h|$ to within $O(1)$, uniformly in the direction.

7.2 The two-mark scalar

For the displacement d, build a cylinder whose period is exactly d: tile the plane by rhombi with side vectors h, v_{lat}, mark the two points 0 and d in a fixed row, and put one marked edge at each. A plane polygon through both v, w is exactly a cylinder polygon through both marked edges once we identify vertices differing by d. Let $\mathscr{H}(d)$ be the total mass of such cylinder polygons, with weight $x_*^{L(\gamma)}$.

Proposition 7.2 (Two-mark bound)**.** *Uniformly in k and in the direction of d, for fixed $C_0 \ge 1$ and $3k \le N \le k^{C_0}$ even, if $k = a + b$ with $a, b > 0$ and $b \le a$, then*

$$|\mathscr{H}(d)| \le k^{-4/3+o(1)}. \tag{23}$$

The exponent $4/3$ is the analogue of the one-mark exponent $2/3$ and comes from two *slope jumps of $3/2$ (one per marked edge).*

The proof of Proposition 7.2 is the two-mark analog of §5–5.6. It has the same three-step structure (angular, radial, spectral) but with a richer charge space because the two distinguished edges break the rotation symmetry that let us simplify in the one-mark case. Section 7.4 sketches the argument. This subsection only uses its conclusion.

7.3 Assembly of the second moment

With Proposition 7.2 in hand, the second moment follows immediately:

$$\begin{aligned}\sum_{D(\gamma)\le r} L(\gamma)^2 w(\gamma) &\le 36 \sum_{\substack{d=a(h-v_{\mathrm{lat}})+bh \\ 0<|d|\le r}} |\mathscr{H}(d)| \\ &\le 36 \sum_{\substack{0<|d|\le r \\ 0<a,b \text{ int}}} k^{-4/3+o(1)}.\end{aligned}$$

The integer pairs (a, b) with $a, b > 0$ and $a + b \le Kr$ for a fixed constant K (from $|d| \le r$) number at most Kr; each has $k = a + b \le Kr$ and contributes $k^{-4/3+o(1)}$. Splitting into dyadic ranges $k \in [K_j, 2K_j)$, there are $O(K_j)$ pairs at scale K_j; the contribution of that scale is at most $K_j \cdot K_j^{-4/3+o(1)} = K_j^{-1/3+o(1)}$. Summing over $K_j = 1, 2, 4, \ldots, r$,

$$\sum_{D(\gamma)\le r} L(\gamma)^2 w(\gamma) \le 36 \sum_{\text{dyadic } K_j \le Kr} K_j^{-1/3+o(1)} \le C\, r^{2/3+o(1)},$$

using the convergence of the geometric series in $K_j^{-1/3}$. This is (22).

7.4 Sketch of the two-mark bound

The two-mark bound has the same three-step structure as the one-mark ratio, but with a doubled sector geometry. We state the seven pieces; each is a finite or analytic calculation that we do not reproduce here in full, but the result and mechanism are explicit.

(a) The scalar identity. From the two-color-block decomposition, the two-mark scalar is an explicit quotient

$$\mathscr{H} = \mathscr{N}/\widetilde{D}(X)^2,$$

where $\widetilde{D}(X) = D_1(X)\prod x_i^{-(N-1)/2}$ is the normalization of the $p = 1$ Pfaffian vacuum (Proposition 2.3), and $\mathscr{N}$ is a doubly-colored block sum over the two blocks R, L. On each block, the operator $M_{p',p'}$ is triangular: at the first change between input and output, the output-minus-input spin equals the change in the intermediate auxiliary spin minus p' (charge conservation). Actually the eigenvalues are products of three basic spins as we run through the three color classes $-, 0, +$. The residual block-charge constraint is exactly $\sum s_L = \sum s_R$, which joins the two blocks. Fully-occupied color terms contract to saturated amplitudes; vacant colors are restored through the fusion identity $D_1(Y, q^{-1}x, qx) = \sqrt{2}\, x \prod_{t\in Y}(t + x)D_1(Y, x)$, giving a color-0 weight of 1 after the constants are stripped. This is the two-mark analog of the identity in §2.5.

(b) Angular algebra. The charge space is the same D_4 of §5.1. Because the two marks are at unequal blocks, the conjugation

$$g^{-1}\widehat{C}\, g = U \cdot D_1 D_2 \widehat{A}$$

now splits into two sectors, each carrying an $\mathfrak{sl}_4$ (the labels 1234 plus the extra root E), and the two sectors commute by nonnegative cross pairings. The rank-one specials $B(Q^3) \propto CC^t$ and $B(Q^2) \propto AA^t$ reduce sector-internally, using the same Weyl phases for the two $\widehat{S}_i$ and the crossing values $B(1) = I$. This is the analog of §2.5 with one extra E root added to the 123 algebra of each sector.

(c) Confluence. The angular trace $\mathcal{T}_\epsilon(L, u_*)$ with the two marks at unequal blocks has the confluent limit

$$\lim_{u_*\to\infty}\lim_{L\to\infty} |\mathcal{T}_\epsilon(L, u_*)| \asymp |\mathscr{N}|(2+\sqrt{2})^N\, M(H, 0)^{N/2} \prod_{j<k} M(H, e_j - e_k),$$

where H is the arithmetic datum of the two blocks and $M(D, r) = d'^{-D_{-r}} \prod_{0\le l<d'} |1 - e^{2\pi i(l+r)/d'}|^{D_l}$ is the C-sine normalization. Each factor is bounded above and below independently of N, of the block sizes, and of the two common shifts of the marks. This is the two-mark analog of the homogeneous-factor formula (17), with the additional $M(H, e_j - e_k)$ pair factors because the two blocks now sit at unequal horizontal coordinates.

(d) Pfaffian normalization. Proposition 4.1 gives $|D_1(X)| = P^{N^2} b^N N^{-7/12+o(1)}$ for each block. Squaring and normalizing, $|\widetilde{D}(X)|^2 = N^{-7/6+o(1)}$ in the denominator of $\mathscr{H}$. The numerator $\mathscr{N}$ has the same $P^{N^2} b^N$-algebraic scale as a product; the huge algebraic factor cancels between numerator and denominator, leaving the subexponential $N^{-7/6}$ in the denominator and a specific colored numerator.

(e) Coercivity. The radial bound (21) applies to each sector separately; the doubled system inherits the same coercive minorant, uniform in the block ratio $\rho = a_2/a_1$ (with $a_1 \asymp N$, $a_2 \asymp k$).

(f) Partial transpose. Partial transposition in the shorter sector converts the four rank-4 roots of D_4 into trace identities with specific exponential rates. The relevant invariant factors are (using $p_Q(r) = \mathfrak{p}(Q_L^r)$ with $\mathfrak{p}(y) = \prod_{m\geq 1}(1-y^m)$)

$$Z_{B'} = p_Q(1)p_Q(6)/p_Q(2), \qquad Z_C = p_Q(1)/p_Q(8), \qquad Z_W = p_Q(1)/p_Q(3), \qquad Z_{B_4} = p_Q(1)^2/p_Q(3)p_Q(6),$$

each via the Euler-product transformation by a cyclotomic factor $\Phi_j(x)$ which contributes $\sum_{d|j} m_d/(24d)$ to the exponential rate. The total rates are

$$Z_{B'} : -\frac{1}{192}, \qquad Z_C : -\frac{1}{96}, \qquad Z_W : \frac{1}{18}, \qquad Z_{B_4} : \frac{1}{144}.$$

These are the two-mark analogs of the Z_R^r factor in §5.3.

(g) Hierarchical assembly. The two sectors have slope labeled by $s_L \in S_0$ (sector L) and $s_R \in S_1$ (sector R). At each mark the slope must jump by exactly $3/2$ (one per marked edge), forcing $s_R \geq s_L + 3/2$. The heat cost per box of width D is $e^{-D(s_L^2+s_R^2)/2}$, the sector-prefactor contributes $2/32$ in both sectors, and the partial-transpose rates $Z_{B'}, Z_C, Z_W, Z_{B_4}$ combine linearly. The retained spectral labels give a per-box rate at most

$$\frac{1}{16} - \frac{1}{4} = -\frac{3}{16},$$

as in the one-mark case. Over the shorter block of length k (which controls the diagonal of the transfer), the shorter-block length matches k boxes; the Pfaffian denominator of part (d) contributes $N^{-7/6}$; and the two-mark structure produces the total

$$|\mathscr{H}| \leq k^{-4/3+o(1)}.$$

The exponent $4/3$ is not a coincidence: it is twice the one-mark exponent $2/3$, corresponding to two slope jumps of $3/2$. The finite spectral enumeration that fixes it is the same one used in §5.6, with two sectors instead of one.

This completes the sketch, and with Proposition 7.2 the second moment (22) follows.

Remark. What is genuinely new in this section relative to §5 is the two-sector structure: the $\mathfrak{sl}_4$ conjugation of part (b), the confluence $M(H, e_j - e_k)$ pair factors of part (c), and the partial-transpose rates of part (f). All three are exact finite computations at homogeneous sites; none of them requires positivity beyond the coercivity of part (e). The reader who wants every matrix entry can consult the original paper's Section 7; here we have condensed to the structural skeleton.

8 Boundary paths and guide corridors

Up to now everything has been about closed polygons. The same transfer machinery controls *open* paths on the honeycomb, with pinned boundary endpoints; this section states the two main boundary results.

8.1 Half-plane arches

Consider the honeycomb upper half-plane cut along row lines. For a fixed boundary port o (an edge midpoint on the boundary line), let h_n be the total critical weight $x_*^{L(\gamma)}$ of self-avoiding walks γ from o to the port n spacings to its right, staying in the closed upper half-plane above the row cut.

Theorem 8.1 (Flat boundary arches)**.** *Let* $h_*^{-1} = 2\cos(3\pi/8)$*. Then*

$$\sum_{n\geq 1} h_n = h_*, \qquad c\, n^{-5/4} \leq h_n \leq C\, n^{-5/4} \quad (n \geq 1), \tag{24}$$

and $h_n[\operatorname{diam}\gamma \leq Cn] \geq cn^{-5/4}$*. The total mass from a fixed port to any other boundary port, restricted to paths of diameter greater than* $R \geq 1$*, is at most* $CR^{-1/4}$ *and at least* $cR^{-1/4}$*.*

Proof sketch. The first ingredient is a boundary analog of the finite polygon identity: a three-root parafermionic identity (Duminil-Copin–Smirnov with spin 5/8) gives, for a bounded convex domain cut along row lines,

$$\sum_p (p - V)F(p) = 0$$

at every interior vertex V with neighbouring midpoints as complex coordinates; $F(p)$ is the sum of $x_*^{\#\text{vertices}} \exp(-i\sigma\,\text{wind})$ over self-avoiding walks from o to p, $\sigma = 5/8$. The identity is local; summing it against a perfect matchings of roots in the four boundary intervals around two fixed roots a, c gives a matrix equation for the four "deficit" sums M_{ij} (deficits in the two-boundary-interval decomposition), with an error term equal to a point-to-point bridge weight $H(a, c)$; the coefficients are explicit powers of $e^{i\pi/16}$ and satisfy a rank-one compatibility condition by the arithmetic of $\mathbb{Q}(e^{i\pi/8})$.

On a flat half-plane, three roots suffice. The pointwise decay $h_n \asymp n^{-5/4}$ comes from the three-root identity at $0, n, n+1$, together with monotonicity (from the two-root cross-identity) and the extremal evaluation identity of the wedge arch mass; the flat total $h_* = 1/(2\cos(3\pi/8))$ follows by saturating the half-plane at the boundary root and taking a limit.

The tail bound is a separate computation: given a fixed port and diameter $> R$, decompose the arch into a stay-in-a-normal-tube part plus an escape. A normal tube of width a small constant times R stays in the half-plane forever; the two-root parafermionic identities bound the opposite-side exit by $cR^{-1/4}$. □

8.2 Slab corridors

The triangle of results about slabs (a strip between parallel cuts) is the analogue of Theorem 8.1 for bridges.

Theorem 8.2 (Slab corridors)**.** *Fix a bottom port in the lower cut of a slab between two parallel row cuts at distance* h *(measured in bond spacings). Let* B_h *be the total critical bridge mass ending at a port of the upper cut. Then*

$$c\, h^{-1/4} \leq B_h \leq C\, h^{-1/4}. \tag{25}$$

Moreover, a positive fraction $c_1 h^{-1/4}$ *of* B_h *lies in the prescribed narrow corridor of width* $c_2 h$ *tilted at any given supported slope, and the mass outside every such corridor is bounded by* $c_3 h^{-1/4}$*.*

Proof sketch. The upper bound $B_h \leq Ch^{-1/4}$ is a single-identity comparison: take the flat half-plane decomposition and subtract the same-side arches lost by restriction to the slab. There are two losses, one at the bottom (arches entering but not leaving) and one at the top; both are positive and their sum bounds B_h.

The lower bound uses the wedge. In an upward-open wedge of angle $2\pi/3$ with ports a, c at distance comparable to k from the apex on opposite sides, the boundary identity gives cross-mass $\asymp k^{-1/4}$. Summing over k-neighbourhoods, this gives $B_h \geq ch^{-1/4}$. The corridor concentration is a finer version: the mass of bridges inside a narrow tilted corridor is a positive fraction of B_h when

the tilted cut is at the appropriate angle, and the assembly uses fairly many slopes iteratively to reach a polygonal guide.

Open paths with fixed nearby endpoints. The same transfer identities extend to fixed-endpoint half-plane walks: the cap $P_{0,0}, R_{0,-1}, P_{1,-1}, R_{1,-1}, P_{1,0}$ injects such walks into plane polygons with a fixed lowest vertex, and the reverse injection is by toggling a hexagonal face. This gives a bijection with polygon classes and transfers the length-moment bound $r^{2/3+o(1)}$ to the half-plane walk, yielding length cutoff exponent $-3/2 + o(1)$ under the half-plane edge weight. □

Remark. The boundary results are logically independent of the closed-polygon results: Theorems 8.1 and 8.2 can be read after §2.1–§2.2, without the radial spectral machinery. The two are needed only for the second-moment bound of §7, where the injection of §8 transfers the length-moment bound to open paths.

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